

insert on-CPU 콜체인 — config 병렬 PoC (2026-06-01)

cub_server testdb 단일 프로세스 on-CPU 캡처 (-F 999, dwarf call-graph) 에서 locator_insert_force 를 root 로 한 콜체인. funcset 제한 없이 script 에 등장한 모든 심볼이 후보 (커널 [unknown] 프레임만 제외).

캡처 정보

항목	값
디렉토리	20260601-203144-cubserver-oncpu-config-parallel-poc-insert/
전체 sample	147,236 (symbol 파싱된 stack)
locator_insert_force 포함 sample	43,840 (전체의 29.78%) = 콜트리 분모
콜체인 함수 수	529개 (root 경로 inclusive)
진입 경로	복제 적용 경로 (slocator_repl_force → xlocator_repl_force → ... → locator_insert_force). 클라이언트 SQL 엔트리 qexec_execute_insert = 0

※ 노드 % = root(=locator_insert_force) 포함 sample 대비 inclusive 비율. 한 함수가 여러 호출 지점에 나타나면 그 전역 inclusive %가 매 위치마다 동일하게 찍힌다(master REPORT 와 동일 규칙). 그래서 자식 %가 부모 %보다 커 보일 수 있음(예: log_append_undoredo_crumbs 27.75%).

설정

항목	값
double_write_buffer_size	0
data_buffer_size	5G
log_buffer_size	5G
log_volume_size	1G
checkpoint_interval	30min
csql>	checkpoint 수행
addvoldb	100G
addvoldb	temp

상위 inclusive 함수 (top 25 / 529)

% of root	samples	함수	의미
100.00	43,840	locator_insert_force	insert force (root)
86.15	37,769	heap_insert_logical	heap insert
34.38	15,070	heap_get_insert_location_with_lock	insert 위치+lock
33.83	14,831	heap_stats_find_best_page	free space 탐색
27.75	12,167	log_append_undoredo_crumbs	undo/redo 로그 append
23.13	10,139	heap_vpid_alloc	페이지 할당
20.90	9,163	lock_object	오브젝트 lock
19.44	8,523	lock_internal_perform_lock_object	lock 수행
18.15	7,957	file_alloc	파일 할당

% of root	samples	함수	의미
17.67	7,747	heap_log_insert_physical	heap 물리 로깅
15.04	6,593	prior_lsa_next_record_internal	prior LSA 다음 레코드
14.87	6,517	__GI___libc_malloc	malloc
13.96	6,118	prior_lsa_alloc_and_copy_crumbs	prior LSA 레코드 생성
12.33	5,406	lockfree_hashmap<lk_res_key, lk_res>::find_or_insert	lock 해시맵
12.26	5,373	_int_malloc	malloc 내부
11.96	5,241	lf_hash_insert_internal	lock-free hash insert
11.65	5,108	lf_list_insert_internal	lock-free list insert
11.55	5,062	cub_alloc	할당 wrapper
11.07	4,851	__GI___pthread_mutex_lock	mutex lock
10.70	4,690	spage_insert_at	slotted page insert
9.97	4,370	locator_add_or_remove_index_internal	인덱스 갱신
9.86	4,321	prior_lsa_gen_undoredo_record_from_crumbs	prior LSA 레코드 조립
9.68	4,245	__memmove_evex_unaligned_erms	memmove
9.37	4,106	sysmalloc	OS 메모리 확장

전체 529개 함수는 아래 §"전체 함수 inclusive 표" 참고.

관찰 (insert)

- write 무게중심: heap_insert_logical (86%) 안에서 **(1) lock 경로**
 heap_get_insert_location_with_lock → lock_object → lock-free hashmap(lf_*, ~12%), **(2) 로그 경로**
 log_append_undoredo_crumbs → prior_lsa_alloc_and_copy_crumbs/prior_lsa_next_record_internal (~14~15%),
(3) free space heap_stats_find_best_page → heap_vpid_alloc → file_alloc (~18~34%) 세 갈래.
- prior_lsa_* + log_append_* 경로가 두껍게 잡힘 → 로그 생성 비용이 insert on-CPU 의 큰 축.
- malloc 체인(cub_alloc → __GI___libc_malloc → _int_malloc → sysmalloc → __mprotect 8.25%) 이 prior_lsa 레코드 복사에 매달려 반복 등장.

콜트리 (locator_insert_force = 100%, 노드컷 0.1%, MIN_EDGE 30)

```
# locator_insert_force 콜체인 (ROOT 포함 sample 만, ROOT = 100%)
# root_samples = 43840 (total_samples = 147236)
# 노드 컷 = 0.1% of root, MIN_EDGE = 30

[100.00%] locator_insert_force
├─ [ 86.15%] heap_insert_logical
│   └─ [ 34.38%] heap_get_insert_location_with_lock
│       └─ [ 33.83%] heap_stats_find_best_page
│           └─ [ 23.13%] heap_vpid_alloc
│               └─ [ 18.15%] file_alloc
│                   └─ [ 5.84%] pgbuf_fix_release
│                       └─ [ 5.14%] pgbuf_claim_bcb_for_fix
│                           └─ [ 4.39%] fileio_init_lsa_of_page
│                               └─ [ 4.46%] LSA_SET_NULL
│                                   └─ [ 0.43%] pgbuf_allocate_bcb
│                                       └─ [ 0.34%] pgbuf_get_bcb_from_invalid_list
│                                           └─ [ 0.10%] pgbuf_lock_page
│                                               └─ [ 0.13%] pgbuf_search_hash_chain
```

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└─ [ 4.16%] file_perm_alloc
└─ [ 2.01%] log_append_undoredo_data2
└─ [ 27.75%] log_append_undoredo_crumbs
└─ [ 13.96%] prior_lsa_alloc_and_copy_crumbs
└─ [ 9.86%] prior_lsa_gen_undoredo_record_from_crumbs
└─ [ 4.42%] log_zip
└─ [ 3.50%] LZ4_resetStream_fast
└─ [ 3.19%] __memset_evex_unaligned_erms
└─ [ 0.26%] __tls_get_addr
└─ [ 11.55%] cub_alloc
└─ [ 14.87%] __GI__libc_malloc
└─ [ 12.26%] _int_malloc
└─ [ 9.37%] sysmalloc
└─ [ 8.25%] __GI__mprotect
└─ [ 0.30%] unlink_chunk.isra.2
└─ [ 9.68%] __memmove_evex_unaligned_erms
└─ [ 0.51%] prior_lsa_copy_redo_crumbs_to_node
└─ [ 11.55%] cub_alloc
└─ [ 14.87%] __GI__libc_malloc
└─ [ 12.26%] _int_malloc
└─ [ 9.37%] sysmalloc
└─ [ 8.25%] __GI__mprotect
└─ [ 0.30%] unlink_chunk.isra.2
└─ [ 0.41%] prior_lsa_copy_undo_crumbs_to_node
└─ [ 11.55%] cub_alloc
└─ [ 14.87%] __GI__libc_malloc
└─ [ 12.26%] _int_malloc
└─ [ 9.37%] sysmalloc
└─ [ 8.25%] __GI__mprotect
└─ [ 0.30%] unlink_chunk.isra.2
└─ [ 0.35%] log_diff
└─ [ 0.30%] LOG_FIND_CURRENT_TDES
└─ [ 3.30%] LOG_FIND_TDES
└─ [ 0.13%] log_zip@plt
└─ [ 0.12%] pgbuf_get_vpid_ptr
└─ [ 11.55%] cub_alloc
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└─ [ 12.26%] _int_malloc
└─ [ 9.37%] sysmalloc
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└─ [ 0.30%] unlink_chunk.isra.2
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└─ [ 11.55%] cub_alloc
└─ [ 14.87%] __GI__libc_malloc
└─ [ 12.26%] _int_malloc

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└ [ 9.37%] sysmalloc
  └ [ 8.25%] __GI__mprotect
    └ [ 0.30%] unlink_chunk.isra.2
└ [ 15.04%] prior_lsa_next_record_internal
  └ [ 11.07%] __GI__pthread_mutex_lock
    └ [ 5.34%] __lll_lock_wait
  └ [ 7.75%] __pthread_mutex_unlock_usercnt
    └ [ 4.29%] __lll_unlock_wake
  └ [ 0.45%] prior_lsa_start_append
    └ [ 0.14%] log_tdes::is_system_worker_transaction
  └ [ 0.37%] vacuum_get_log_blockid
  └ [ 0.34%] __gthread_mutex_unlock
  └ [ 0.25%] prior_lsa_append_data
    └ [ 0.12%] log_prior_lsa_append_align
  └ [ 0.25%] log_prior_lsa_append_advance_when_doesnot_fit
    └ [ 0.13%] logpb_get_memsize
└ [ 0.39%] pgbuf_set_lsa
  └ [ 0.50%] pgbuf_set_dirty_buffer_ptr
    └ [ 0.34%] pgbuf_find_thrd_holder
      └ [ 0.48%] pgbuf_set_dirty
└ [ 3.30%] LOG_FIND_TDES
└ [ 0.52%] logtb_find_client_type
  └ [ 3.30%] LOG_FIND_TDES
└ [ 0.27%] pgbuf_is_lsa_temporary
  └ [ 0.22%] xdisk_get_purpose
    └ [ 0.12%] log_can_skip_undo_logging
└ [ 9.27%] log_append_undoredo_data
  └ [ 27.75%] log_append_undoredo_crumbs
    └ [ 13.96%] prior_lsa_alloc_and_copy_crumbs
      └ [ 9.86%] prior_lsa_gen_undoredo_record_from_crumbs
        └ [ 4.42%] log_zip
          └ [ 3.50%] LZ4_resetStream_fast
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                      └ [ 0.30%] unlink_chunk.isra.2
          └ [ 0.41%] prior_lsa_copy_undo_crumbs_to_node
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              └ [ 14.87%] __GI__libc_malloc
                └ [ 12.26%] _int_malloc
                  └ [ 9.37%] sysmalloc
                    └ [ 8.25%] __GI__mprotect
                      └ [ 0.30%] unlink_chunk.isra.2
          └ [ 0.35%] log_diff
          └ [ 0.30%] LOG_FIND_CURRENT_TDES

```

```

└─ [ 3.30%] LOG_FIND_TDES
└─ [ 0.13%] log_zip@plt
└─ [ 0.12%] pgbuf_get_vpid_ptr
└─ [ 11.55%] cub_alloc
└─ [ 14.87%] __GI__libc_malloc
└─ [ 12.26%] _int_malloc
└─ [ 9.37%] sysmalloc
└─ [ 8.25%] __GI__mprotect
└─ [ 0.30%] unlink_chunk.isra.2
└─ [ 1.22%] prior_lsa_copy_redo_data_to_node
└─ [ 11.55%] cub_alloc
└─ [ 14.87%] __GI__libc_malloc
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└─ [ 0.30%] unlink_chunk.isra.2
└─ [ 15.04%] prior_lsa_next_record_internal
└─ [ 11.07%] __GI__pthread_mutex_lock
└─ [ 5.34%] __lll_lock_wait
└─ [ 7.75%] __pthread_mutex_unlock_usercnt
└─ [ 4.29%] __lll_unlock_wake
└─ [ 0.45%] prior_lsa_start_append
└─ [ 0.14%] log_tdes::is_system_worker_transaction
└─ [ 0.37%] vacuum_get_log_blockid
└─ [ 0.34%] __gthread_mutex_unlock
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└─ [ 0.12%] log_prior_lsa_append_align
└─ [ 0.25%] log_prior_lsa_append_advance_when_doesnot_fit
└─ [ 0.13%] logpb_get_memszie
└─ [ 0.39%] pgbuf_set_lsa
└─ [ 0.50%] pgbuf_set_dirty_buffer_ptr
└─ [ 0.34%] pgbuf_find_thrd_holder
└─ [ 0.48%] pgbuf_set_dirty
└─ [ 3.30%] LOG_FIND_TDES
└─ [ 0.52%] logtb_find_client_type
└─ [ 3.30%] LOG_FIND_TDES
└─ [ 0.27%] pgbuf_is_lsa_temporary
└─ [ 0.22%] xdisk_get_purpose
└─ [ 0.12%] log_can_skip_undo_logging
└─ [ 2.96%] heap_vpid_init_new
└─ [ 9.27%] log_append_undoredo_data
└─ [ 27.75%] log_append_undoredo_crumbs
└─ [ 13.96%] prior_lsa_alloc_and_copy_crumbs
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└─ [ 4.42%] log_zip

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└ [ 3.50%] LZ4_resetStream_fast
  └ [ 3.19%] __memset_evex_unaligned_erms
    └ [ 0.26%] __tls_get_addr
└ [ 11.55%] cub_alloc
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          └ [ 0.30%] unlink_chunk.isra.2
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└ [ 0.51%] prior_lsa_copy_redo_crumbs_to_node
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          └ [ 8.25%] __GI__mprotect
            └ [ 0.30%] unlink_chunk.isra.2
└ [ 15.04%] prior_lsa_next_record_internal
  └ [ 11.07%] __GI__pthread_mutex_lock
    └ [ 5.34%] __l1l_lock_wait

```

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└─ [ 7.75%] __pthread_mutex_unlock_usercnt
    │
    │   └─ [ 4.29%] __lll_unlock_wake
    │
    │   └─ [ 0.45%] prior_lsa_start_append
    │       │
    │       │   └─ [ 0.14%] log_tdes::is_system_worker_transaction
    │       │
    │       │   └─ [ 0.37%] vacuum_get_log_blockid
    │       │
    │       │   └─ [ 0.34%] __gthread_mutex_unlock
    │       │
    │       │   └─ [ 0.25%] prior_lsa_append_data
    │       │       │
    │       │       │   └─ [ 0.12%] log_prior_lsa_append_align
    │       │       │
    │       │       │   └─ [ 0.25%] log_prior_lsa_append_advance_when_doesnot_fit
    │       │       │       │
    │       │       │       │   └─ [ 0.13%] logpb_get_memsize
    │       │       │
    │       │       └─ [ 0.39%] pgbuf_set_lsa
    │       │           │
    │       │           │   └─ [ 0.50%] pgbuf_set_dirty_buffer_ptr
    │       │           │       │
    │       │           │       │   └─ [ 0.34%] pgbuf_find_thrd_holder
    │       │           │       │       │
    │       │           │       │       │   └─ [ 0.48%] pgbuf_set_dirty
    │       │       └─ [ 3.30%] LOG_FIND_TDES
    │       └─ [ 0.52%] logtb_find_client_type
    │           │
    │           │   └─ [ 3.30%] LOG_FIND_TDES
    │       └─ [ 0.27%] pgbuf_is_lsa_temporary
    │           │
    │           │   └─ [ 0.22%] xdisk_get_purpose
    │       └─ [ 0.12%] log_can_skip_undo_logging
    └─ [ 0.76%] spage_insert
        │
        │   └─ [ 0.65%] spage_find_empty_slot
        │       │
        │       │   └─ [ 1.51%] spage_has_enough_total_space
        │       │       │
        │       │       │   └─ [ 2.01%] spage_get_total_saved_spaces
        │       │       │       │
        │       │       │       │   └─ [ 1.93%] spage_get_saved_spaces
        │       │       │       │       │
        │       │       │       │       │   └─ [ 1.48%] lf_hash_find
        │       │       │       │       │       │
        │       │       │       │       │       │   └─ [ 0.87%] lf_list_find
        │       │       │       │       │       │       │
        │       │       │       │       │       │       │   └─ [ 0.19%] lf_callback_vpid_hash
        │       │       │       │       │       │       │       │
        │       │       │       │       │       │       │       │   └─ [ 0.39%] logtb_find_tranid
        │       │       │       │       │       │       │       │       │
        │       │       │       │       │       │       │       │       │   └─ [ 3.30%] LOG_FIND_TDES
        │       │       │       │       │       │       │
        │       │       │       │       │       │       └─ [ 0.39%] logtb_find_tranid
        │       │       │       │       │       └─ [ 3.30%] LOG_FIND_TDES
        │       │       └─ [ 1.13%] spage_check_space
        │       │           │
        │       │           │   └─ [ 1.51%] spage_has_enough_total_space
        │       │           │       │
        │       │           │       │   └─ [ 2.01%] spage_get_total_saved_spaces
        │       │           │       │       │
        │       │           │       │       │   └─ [ 1.93%] spage_get_saved_spaces
        │       │           │       │       │       │
        │       │           │       │       │       │   └─ [ 1.48%] lf_hash_find
        │       │           │       │       │       │       │
        │       │           │       │       │       │       │   └─ [ 0.87%] lf_list_find
        │       │           │       │       │       │       │       │
        │       │           │       │       │       │       │       │   └─ [ 0.19%] lf_callback_vpid_hash
        │       │           │       │       │       │       │       │       │
        │       │           │       │       │       │       │       │       │   └─ [ 0.39%] logtb_find_tranid
        │       │           │       │       │       │       │       │       │       │
        │       │           │       │       │       │       │       │       │       │   └─ [ 3.30%] LOG_FIND_TDES
        │       │           │       │       │       │       │
        │       │           │       │       │       │       └─ [ 0.39%] logtb_find_tranid
        │       │           │       │       │       └─ [ 3.30%] LOG_FIND_TDES
        │       └─ [ 0.15%] spage_initialize
        └─ [ 0.48%] pgbuf_set_dirty
            │
            │   └─ [ 0.50%] pgbuf_set_dirty_buffer_ptr
            │       │
            │       │   └─ [ 0.34%] pgbuf_find_thrd_holder
        └─ [ 2.13%] log_sysop_end_logical_undo
            │
            │   └─ [ 3.85%] log_sysop_commit_internal
            │       │
            │       │   └─ [ 3.29%] log_append_sysop_end
            │       │       │
            │       │       │   └─ [ 15.04%] prior_lsa_next_record_internal
            │       │       │       │
            │       │       │       │   └─ [ 11.07%] __GI__pthread_mutex_lock
            │       │       │       │       │
            │       │       │       │       │   └─ [ 5.34%] __lll_lock_wait
            │       │       │       │       │       │
            │       │       │       │       │       │   └─ [ 7.75%] __pthread_mutex_unlock_usercnt
            │       │       │       │       │       │       │
            │       │       │       │       │       │       │   └─ [ 4.29%] __lll_unlock_wake
            │       │       │       │       │       │       │       │
            │       │       │       │       │       │       │       │   └─ [ 0.45%] prior_lsa_start_append
            │       │       │       │       │       │       │       │       │
            │       │       │       │       │       │       │       │       │   └─ [ 0.14%] log_tdes::is_system_worker_transaction
    
```

```

└─ [ 0.37%] vacuum_get_log_blockid
└─ [ 0.34%] __gthread_mutex_unlock
└─ [ 0.25%] prior_lsa_append_data
└─ [ 0.12%] log_prior_lsa_append_align
└─ [ 0.25%] log_prior_lsa_append_advance_when_doesnot_fit
└─ [ 0.13%] logpb_get_memsized
└─ [ 1.49%] prior_lsa_alloc_and_copy_data
└─ [ 11.55%] cub_alloc
└─ [ 14.87%] __GI__libc_malloc
└─ [ 12.26%] _int_malloc
└─ [ 9.37%] sysmalloc
└─ [ 8.25%] __GI__mprotect
└─ [ 0.30%] unlink_chunk.isra.2
└─ [ 0.47%] prior_lsa_gen_record
└─ [ 11.55%] cub_alloc
└─ [ 14.87%] __GI__libc_malloc
└─ [ 12.26%] _int_malloc
└─ [ 9.37%] sysmalloc
└─ [ 8.25%] __GI__mprotect
└─ [ 0.30%] unlink_chunk.isra.2
└─ [ 0.33%] prior_lsa_copy_undo_data_to_node
└─ [ 11.55%] cub_alloc
└─ [ 14.87%] __GI__libc_malloc
└─ [ 12.26%] _int_malloc
└─ [ 9.37%] sysmalloc
└─ [ 8.25%] __GI__mprotect
└─ [ 0.30%] unlink_chunk.isra.2
└─ [ 0.28%] log_tdes::unlock_topop
└─ [ 0.29%] cubpl::get_session
└─ [ 0.21%] session_get_pl_session
└─ [ 0.13%] cubpl::session::is_sp_running
└─ [ 0.10%] log_sysop_end_final
└─ [ 15.04%] prior_lsa_next_record_internal
└─ [ 11.07%] __GI__pthread_mutex_lock
└─ [ 5.34%] __lll_lock_wait
└─ [ 7.75%] __pthread_mutex_unlock_usercnt
└─ [ 4.29%] __lll_unlock_wake
└─ [ 0.45%] prior_lsa_start_append
└─ [ 0.14%] log_tdes::is_system_worker_transaction
└─ [ 0.37%] vacuum_get_log_blockid
└─ [ 0.34%] __gthread_mutex_unlock
└─ [ 0.25%] prior_lsa_append_data
└─ [ 0.12%] log_prior_lsa_append_align
└─ [ 0.25%] log_prior_lsa_append_advance_when_doesnot_fit
└─ [ 0.13%] logpb_get_memsized
└─ [ 4.38%] pgbuf_unfix
└─ [ 2.70%] pgbuf_unlatch_bcb_upon_unfix
└─ [ 0.60%] pgbuf_unlatch_void_zone_bcb
└─ [ 0.41%] pgbuf_lru_add_new_bcb_to_top
└─ [ 0.32%] pgbuf_lru_adjust_zones
└─ [ 0.13%] pgbuf_lru_adjust_zone1
└─ [ 0.21%] pgbuf_bcb_change_zone
└─ [ 0.20%] ATOMIC_CAS_32<int, int, int>
└─ [ 0.13%] pgbuf_lru_fall_bcb_to_zone_3
└─ [ 0.21%] pgbuf_bcb_register_hit_for_lru
└─ [ 11.07%] __GI__pthread_mutex_lock
└─ [ 5.34%] __lll_lock_wait

```



```

└─ [ 7.75%] pthread_mutex_unlock_usercnt
    └─ [ 4.29%] __l1l_unlock_wake
└─ [ 0.25%] pgbuf_wakeup_reader_writer
    └─ [ 0.25%] set_waiter_exists
        └─ [ 0.21%] pgbuf_bcb_register_hit_for_lru
└─ [ 11.07%] __GI__pthread_mutex_lock
    └─ [ 5.34%] __l1l_lock_wait
└─ [ 0.38%] pgbuf_unlatch_thrd_holder
    └─ [ 0.19%] pgbuf_remove_thrd_holder
        └─ [ 0.34%] pgbuf_find_thrd_holder
└─ [ 0.17%] perfmon_is_perf_tracking
    └─ [ 0.22%] xdisk_get_purpose
└─ [ 0.60%] log_sysop_start_atomic
    └─ [ 1.49%] prior_lsa_alloc_and_copy_data
        └─ [ 11.55%] cub_alloc
            └─ [ 14.87%] __GI__libc_malloc
                └─ [ 12.26%] _int_malloc
                    └─ [ 9.37%] sysmalloc
                        └─ [ 8.25%] __GI__mprotect
                            └─ [ 0.30%] unlink_chunk.isra.2
└─ [ 0.47%] prior_lsa_gen_record
    └─ [ 11.55%] cub_alloc
        └─ [ 14.87%] __GI__libc_malloc
            └─ [ 12.26%] _int_malloc
                └─ [ 9.37%] sysmalloc
                    └─ [ 8.25%] __GI__mprotect
                        └─ [ 0.30%] unlink_chunk.isra.2
└─ [ 0.33%] prior_lsa_copy_undo_data_to_node
    └─ [ 11.55%] cub_alloc
        └─ [ 14.87%] __GI__libc_malloc
            └─ [ 12.26%] _int_malloc
                └─ [ 9.37%] sysmalloc
                    └─ [ 8.25%] __GI__mprotect
                        └─ [ 0.30%] unlink_chunk.isra.2
└─ [ 0.32%] log_sysop_start
    └─ [ 0.20%] log_tdes::lock_topop
        └─ [ 0.29%] cubpl::get_session
            └─ [ 0.21%] session_get_pl_session
                └─ [ 0.13%] cubpl::session::is_sp_running
└─ [ 9.27%] log_append_undoredo_data
    └─ [ 27.75%] log_append_undoredo_crums
        └─ [ 13.96%] prior_lsa_alloc_and_copy_crums
            └─ [ 9.86%] prior_lsa_gen_undoredo_record_from_crums
                └─ [ 4.42%] log_zip
                    └─ [ 3.50%] LZ4_resetStream_fast
                        └─ [ 3.19%] __memset_evex_unaligned_erms
                            └─ [ 0.26%] __tls_get_addr
└─ [ 11.55%] cub_alloc
    └─ [ 14.87%] __GI__libc_malloc
        └─ [ 12.26%] _int_malloc
            └─ [ 9.37%] sysmalloc
                └─ [ 8.25%] __GI__mprotect
                    └─ [ 0.30%] unlink_chunk.isra.2
└─ [ 9.68%] __memmove_evex_unaligned_erms
└─ [ 0.51%] prior_lsa_copy_redo_crums_to_node
    └─ [ 11.55%] cub_alloc
        └─ [ 14.87%] __GI__libc_malloc

```

```

└─ [ 12.26%] _int_malloc
    └─ [ 9.37%] sysmalloc
        └─ [ 8.25%] __GI___mprotect
            └─ [ 0.30%] unlink_chunk.isra.2
└─ [ 0.41%] prior_lsa_copy_undo_crumbs_to_node
    └─ [ 11.55%] cub_alloc
        └─ [ 14.87%] __GI___libc_malloc
            └─ [ 12.26%] _int_malloc
                └─ [ 9.37%] sysmalloc
                    └─ [ 8.25%] __GI___mprotect
                        └─ [ 0.30%] unlink_chunk.isra.2
└─ [ 0.35%] log_diff
└─ [ 0.30%] LOG_FIND_CURRENT_TDES
    └─ [ 3.30%] LOG_FIND_TDES
└─ [ 0.13%] log_zip@plt
    └─ [ 0.12%] pgbuf_get_vpid_ptr
└─ [ 11.55%] cub_alloc
    └─ [ 14.87%] __GI___libc_malloc
        └─ [ 12.26%] _int_malloc
            └─ [ 9.37%] sysmalloc
                └─ [ 8.25%] __GI___mprotect
                    └─ [ 0.30%] unlink_chunk.isra.2
└─ [ 1.22%] prior_lsa_copy_redo_data_to_node
    └─ [ 11.55%] cub_alloc
        └─ [ 14.87%] __GI___libc_malloc
            └─ [ 12.26%] _int_malloc
                └─ [ 9.37%] sysmalloc
                    └─ [ 8.25%] __GI___mprotect
                        └─ [ 0.30%] unlink_chunk.isra.2
└─ [ 9.68%] __memmove_evex_unaligned_erms
└─ [ 4.42%] log_zip
    └─ [ 3.50%] LZ4_resetStream_fast
        └─ [ 3.19%] __memset_evex_unaligned_erms
└─ [ 0.26%] __tls_get_addr
└─ [ 0.33%] prior_lsa_copy_undo_data_to_node
    └─ [ 11.55%] cub_alloc
        └─ [ 14.87%] __GI___libc_malloc
            └─ [ 12.26%] _int_malloc
                └─ [ 9.37%] sysmalloc
                    └─ [ 8.25%] __GI___mprotect
                        └─ [ 0.30%] unlink_chunk.isra.2
└─ [ 15.04%] prior_lsa_next_record_internal
    └─ [ 11.07%] __GI___pthread_mutex_lock
        └─ [ 5.34%] __l1l_lock_wait
└─ [ 7.75%] __pthread_mutex_unlock_usercnt
    └─ [ 4.29%] __l1l_unlock_wake
└─ [ 0.45%] prior_lsa_start_append
    └─ [ 0.14%] log_tdes::is_system_worker_transaction
└─ [ 0.37%] vacuum_get_log_blockid
└─ [ 0.34%] __gthread_mutex_unlock
└─ [ 0.25%] prior_lsa_append_data
    └─ [ 0.12%] log_prior_lsa_append_align
└─ [ 0.25%] log_prior_lsa_append_advance_when_doesnot_fit
    └─ [ 0.13%] logpb_get_memszie
└─ [ 0.39%] pgbuf_set_lsa
    └─ [ 0.50%] pgbuf_set_dirty_buffer_ptr
        └─ [ 0.34%] pgbuf_find_thrd_holder

```

```
|      |    | └─ [ 4.38%] pgbuf_unfix
```

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```
└─ [ 13.96%] prior_lsa_alloc_and_copy_crumbs
    │   └─ [ 9.86%] prior_lsa_gen_undoredo_record_from_crumbs
    │       │   └─ [ 4.42%] log_zip
    │           │   └─ [ 3.50%] LZ4_resetStream_fast
    │               │   └─ [ 3.19%] __memset_evex_unaligned_erms
    │                   └─ [ 0.26%] __tls_get_addr
    │   └─ [ 11.55%] cub_alloc
    │       └─ [ 14.87%] __GI__libc_malloc
    │           └─ [ 12.26%] _int_malloc
    │               └─ [ 9.37%] sysmalloc
    │                   └─ [ 8.25%] __GI__mprotect
    │                       └─ [ 0.30%] unlink_chunk.isra.2
    │   └─ [ 9.68%] __memmove_evex_unaligned_erms
    │   └─ [ 0.51%] prior_lsa_copy_redo_crumbs_to_node
    │       └─ [ 11.55%] cub_alloc
    │           └─ [ 14.87%] __GI__libc_malloc
    │               └─ [ 12.26%] _int_malloc
    │                   └─ [ 9.37%] sysmalloc
    │                       └─ [ 8.25%] __GI__mprotect
    │                           └─ [ 0.30%] unlink_chunk.isra.2
    │   └─ [ 0.41%] prior_lsa_copy_undo_crumbs_to_node
    │       └─ [ 11.55%] cub_alloc
    │           └─ [ 14.87%] __GI__libc_malloc
    │               └─ [ 12.26%] _int_malloc
    │                   └─ [ 9.37%] sysmalloc
    │                       └─ [ 8.25%] __GI__mprotect
    │                           └─ [ 0.30%] unlink_chunk.isra.2
    │   └─ [ 0.35%] log_diff
    │   └─ [ 0.30%] LOG_FIND_CURRENT_TDES
    │       └─ [ 3.30%] LOG_FIND_TDES
    │   └─ [ 0.13%] log_zip@plt
    │       └─ [ 0.12%] pgbuf_get_vpid_ptr
    │   └─ [ 11.55%] cub_alloc
    │       └─ [ 14.87%] __GI__libc_malloc
    │           └─ [ 12.26%] _int_malloc
    │               └─ [ 9.37%] sysmalloc
    │                   └─ [ 8.25%] __GI__mprotect
    │                       └─ [ 0.30%] unlink_chunk.isra.2
    │   └─ [ 1.22%] prior_lsa_copy_redo_data_to_node
    │       └─ [ 11.55%] cub_alloc
    │           └─ [ 14.87%] __GI__libc_malloc
    │               └─ [ 12.26%] _int_malloc
    │                   └─ [ 9.37%] sysmalloc
    │                       └─ [ 8.25%] __GI__mprotect
    │                           └─ [ 0.30%] unlink_chunk.isra.2
    │       └─ [ 9.68%] __memmove_evex_unaligned_erms
    │   └─ [ 4.42%] log_zip
    │       └─ [ 3.50%] LZ4_resetStream_fast
    │           └─ [ 3.19%] __memset_evex_unaligned_erms
    │               └─ [ 0.26%] __tls_get_addr
    │   └─ [ 0.33%] prior_lsa_copy_undo_data_to_node
    │       └─ [ 11.55%] cub_alloc
    │           └─ [ 14.87%] __GI__libc_malloc
    │               └─ [ 12.26%] _int_malloc
    │                   └─ [ 9.37%] sysmalloc
    │                       └─ [ 8.25%] __GI__mprotect
    │                           └─ [ 0.30%] unlink chunk.isra.2
```

```

└─ [ 15.04%] prior_lsa_next_record_internal
    │
    │   └─ [ 11.07%] __GI___pthread_mutex_lock
    │       │
    │       │   └─ [ 5.34%] __lll_lock_wait
    │       │
    │       └─ [ 7.75%] __pthread_mutex_unlock_usercnt
    │           │
    │           │   └─ [ 4.29%] __lll_unlock_wake
    │           │
    │           └─ [ 0.45%] prior_lsa_start_append
    │               │
    │               │   └─ [ 0.14%] log_tdes::is_system_worker_transaction
    │               │
    │               └─ [ 0.37%] vacuum_get_log_blockid
    │                   │
    │                   │   └─ [ 0.34%] __gthread_mutex_unlock
    │                   │
    │                   └─ [ 0.25%] prior_lsa_append_data
    │                       │
    │                       │   └─ [ 0.12%] log_prior_lsa_append_align
    │                       │
    │                       └─ [ 0.25%] log_prior_lsa_append_advance_when_doesnot_fit
    │                           │
    │                           └─ [ 0.13%] logpb_get_memszie
    │
    └─ [ 0.39%] pgbuf_set_lsa
        │
        │   └─ [ 0.50%] pgbuf_set_dirty_buffer_ptr
        │       │
        │       │   └─ [ 0.34%] pgbuf_find_thrd_holder
        │       │
        │       └─ [ 0.48%] pgbuf_set_dirty
    └─ [ 3.30%] LOG_FIND_TDES
        │
        │   └─ [ 0.52%] logtb_find_client_type
        │       │
        │       │   └─ [ 3.30%] LOG_FIND_TDES
        │       │
        │       └─ [ 0.27%] pgbuf_is_lsa_temporary
        │           │
        │           │   └─ [ 0.22%] xdisk_get_purpose
        │           │
        │           └─ [ 0.12%] log_can_skip_undo_logging
    └─ [ 4.38%] pgbuf_unfix
        │
        │   └─ [ 2.70%] pgbuf_unlatch_bcb_upon_unfix
        │       │
        │       │   └─ [ 0.60%] pgbuf_unlatch_void_zone_bcb
        │       │       │
        │       │       │   └─ [ 0.41%] pgbuf_lru_add_new_bcb_to_top
        │       │       │       │
        │       │       │       │   └─ [ 0.32%] pgbuf_lru_adjust_zones
        │       │       │       │       │
        │       │       │       │       │   └─ [ 0.13%] pgbuf_lru_adjust_zone1
        │       │       │       │       │       │
        │       │       │       │       │       │   └─ [ 0.21%] pgbuf_bcb_change_zone
        │       │       │       │       │       │       │
        │       │       │       │       │       │       │   └─ [ 0.20%] ATOMIC_CAS_32<int, int, int>
        │       │       │       │       │       │       │
        │       │       │       │       │       └─ [ 0.13%] pgbuf_lru_fall_bcb_to_zone_3
        │       │       │       └─ [ 0.21%] pgbuf_bcb_register_hit_for_lru
        │       │       └─ [ 11.07%] __GI___pthread_mutex_lock
        │       │           │
        │       │           │   └─ [ 5.34%] __lll_lock_wait
        │       │           │
        │       │       └─ [ 7.75%] __pthread_mutex_unlock_usercnt
        │       │           │
        │       │           │   └─ [ 4.29%] __lll_unlock_wake
        │       │           │
        │       │       └─ [ 0.25%] pgbuf_wakeup_reader_writer
        │       │           │
        │       │           │   └─ [ 0.25%] set_waiter_exists
        │       │           │
        │       │       └─ [ 0.21%] pgbuf_bcb_register_hit_for_lru
    └─ [ 11.07%] __GI___pthread_mutex_lock
        │
        │   └─ [ 5.34%] __lll_lock_wait
    └─ [ 0.38%] pgbuf_unlatch_thrd_holder
        │
        │   └─ [ 0.19%] pgbuf_remove_thrd_holder
        │       │
        │       │   └─ [ 0.34%] pgbuf_find_thrd_holder
    └─ [ 0.17%] perfmon_is_perf_tracking
        │
        │   └─ [ 0.22%] xdisk_get_purpose
    └─ [ 0.57%] mht_get2
    └─ [ 0.39%] spage_get_record
        │
        │   └─ [ 0.31%] spage_find_slot
    └─ [ 0.17%] pgbuf_ordered_set_dirty_and_free
        │
        │   └─ [ 0.48%] pgbuf_set_dirty
        │       │
        │       │   └─ [ 0.50%] pgbuf_set_dirty_buffer_ptr
        │       │
        │       └─ [ 0.34%] pgbuf_find_thrd_holder
    └─ [ 0.10%] spage_get_record@plt
    └─ [ 0.23%] pgbuf_ordered_unfix
        │
        │   └─ [ 0.15%] pgbuf_remove_watcher

```

```

└─ [ 20.90%] lock_object
├─ [ 19.44%] lock_internal_perform_lock_object
├─ [ 12.33%] cubthread::lockfree_hashmap<lk_res_key, lk_res>::find_or_insert
├─ [ 11.95%] lf_hash_insert_internal
├─ [ 11.65%] lf_list_insert_internal
├─ [ 6.46%] lock_res_key_compare
├─ [ 8.50%] lf_freelist_claim
├─ [ 6.76%] lf_freelist_alloc_block
├─ [ 14.87%] __GI___libc_malloc
├─ [ 12.26%] _int_malloc
├─ [ 9.37%] sysmalloc
├─ [ 8.25%] __GI___mprotect
├─ [ 0.30%] unlink_chunk.isra.2
├─ [ 3.09%] lock_alloc_resource
├─ [ 11.55%] cub_alloc
├─ [ 14.87%] __GI___libc_malloc
├─ [ 12.26%] _int_malloc
├─ [ 9.37%] sysmalloc
├─ [ 8.25%] __GI___mprotect
├─ [ 0.30%] unlink_chunk.isra.2
├─ [ 0.83%] lf_stack_pop
├─ [ 0.19%] ATOMIC_CAS_ADDR<void>
├─ [ 0.32%] ATOMIC_INC_32<int, int>
├─ [ 11.07%] __GI___pthread_mutex_lock
├─ [ 5.34%] __lll_lock_wait
├─ [ 8.50%] lf_freelist_claim
├─ [ 6.76%] lf_freelist_alloc_block
├─ [ 14.87%] __GI___libc_malloc
├─ [ 12.26%] _int_malloc
├─ [ 9.37%] sysmalloc
├─ [ 8.25%] __GI___mprotect
├─ [ 0.30%] unlink_chunk.isra.2
├─ [ 3.09%] lock_alloc_resource
├─ [ 11.55%] cub_alloc
├─ [ 14.87%] __GI___libc_malloc
├─ [ 12.26%] _int_malloc
├─ [ 9.37%] sysmalloc
├─ [ 8.25%] __GI___mprotect
├─ [ 0.30%] unlink_chunk.isra.2
├─ [ 0.83%] lf_stack_pop
├─ [ 0.19%] ATOMIC_CAS_ADDR<void>
├─ [ 0.32%] ATOMIC_INC_32<int, int>
├─ [ 0.41%] lock_initialize_entry_as_granted
├─ [ 0.69%] lock_event_set_xasl_id_to_entry
├─ [ 3.30%] LOG_FIND_TDES
├─ [ 0.69%] lock_event_set_xasl_id_to_entry
├─ [ 3.30%] LOG_FIND_TDES
├─ [ 0.64%] lock_find_class_entry
├─ [ 11.07%] __GI___pthread_mutex_lock
├─ [ 5.34%] __lll_lock_wait
├─ [ 7.75%] __pthread_mutex_unlock_usercnt
├─ [ 4.29%] __lll_unlock_wake
├─ [ 0.30%] lock_insert_into_tran_hold_list
├─ [ 11.07%] __GI___pthread_mutex_lock
├─ [ 5.34%] __lll_lock_wait
├─ [ 7.75%] __pthread_mutex_unlock_usercnt
├─ [ 4.29%] __lll_unlock_wake

```

```
| | | └─ [ 0.11%] lock_escalate_if_needed  
| | |   └─ [ 11.07%] __GI___pthread_mutex_lock  
| | |     └─ [ 5.34%] _lll_lock_wait  
| | ── [ 0.61%] lock_get_class_lock  
| |   │├─ [ 11.07%] __GI___pthread_mutex_lock  
| |   ││└─ [ 5.34%] _lll_lock_wait  
| |   │├─ [ 0.10%] logtb_get_current_tran_index  
| |   ││└─ [ 0.18%] thread_get_thread_entry_info  
| |   ││   └─ [ 0.14%] cubthread::get_entry  
| |   ││       └─ [ 0.26%] __tls_get_addr  
| |   │└─ [ 7.75%] __pthread_mutex_unlock_usercnt  
| |         └─ [ 4.29%] _lll_unlock_wake  
| ── [ 0.64%] lock_find_class_entry  
|   │├─ [ 11.07%] __GI___pthread_mutex_lock  
|   ││└─ [ 5.34%] _lll_lock_wait  
|   │└─ [ 7.75%] __pthread_mutex_unlock_usercnt  
|         └─ [ 4.29%] _lll_unlock_wake  
└─ [ 0.15%] logtb_find_wait_msecs  
    └─ [ 3.30%] LOG_FIND_TDES  
  
├─ [ 17.67%] heap_log_insert_physical  
|   │├─ [ 27.75%] log_append_undo redo crumbs  
|   ││├─ [ 13.96%] prior_lsa_alloc_and_copy_crumbs  
|   │││├─ [ 9.86%] prior_lsa_gen_undo redo record_from_crumbs  
|   ││││├─ [ 4.42%] log_zip  
|   │││││├─ [ 3.50%] LZ4_resetStream_fast  
|   │││││└─ [ 3.19%] __memset_evex_unaligned_erms  
|   ││││   └─ [ 0.26%] __tls_get_addr  
|   │││└─ [ 11.55%] cub_alloc  
|   │││   └─ [ 14.87%] __GI__libc_malloc  
|   │││       └─ [ 12.26%] _int_malloc  
|   │││           │├─ [ 9.37%] sysmalloc  
|   │││           │└─ [ 8.25%] __GI__mprotect  
|   │││               └─ [ 0.30%] unlink_chunk.isra.2  
|   ││├─ [ 9.68%] __memmove_evex_unaligned_erms  
|   ││├─ [ 0.51%] prior_lsa_copy_redo crumbs_to_node  
|   ││   └─ [ 11.55%] cub_alloc  
|   ││       └─ [ 14.87%] __GI__libc_malloc  
|   ││           └─ [ 12.26%] _int_malloc  
|   ││               │├─ [ 9.37%] sysmalloc  
|   ││               │└─ [ 8.25%] __GI__mprotect  
|   ││                   └─ [ 0.30%] unlink_chunk.isra.2  
|   │├─ [ 0.41%] prior_lsa_copy_undo crumbs_to_node  
|   │   └─ [ 11.55%] cub_alloc  
|   │       └─ [ 14.87%] __GI__libc_malloc  
|   │           └─ [ 12.26%] _int_malloc  
|   │               │├─ [ 9.37%] sysmalloc  
|   │               │└─ [ 8.25%] __GI__mprotect  
|   │                   └─ [ 0.30%] unlink_chunk.isra.2  
|   │├─ [ 0.35%] log_diff  
|   │├─ [ 0.30%] LOG_FIND_CURRENT_TDES  
|   ││└─ [ 3.30%] LOG_FIND_TDES  
|   │├─ [ 0.13%] log_zip@plt  
|   ││└─ [ 0.12%] pgbuf_get_vpid_ptr  
|   └─ [ 11.55%] cub_alloc  
|       └─ [ 14.87%] __GI__libc_malloc  
|           └─ [ 12.26%] _int_malloc  
|               │├─ [ 9.37%] sysmalloc
```


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```

| | |└ [ 0.29%] heap_attrinfo_recache_attrepr
| | |└ [ 0.11%] db_value_domain_init
| |└ [ 0.16%] hl_lea_alloc
|└ [ 0.31%] btree_insert
| |└ [ 0.11%] btree_mvcc_info_from_heap_mvcc_header
|└ [ 0.78%] logtb_get_current_mvccid
| |└ [ 3.30%] LOG_FIND_TDES
|└ [ 0.51%] mvcc_is_mvcc_disabled_class
|└ [ 0.52%] logtb_find_client_type
| |└ [ 3.30%] LOG_FIND_TDES
|└ [ 0.14%] mvcc_is_mvcc_disabled_class@plt
└ [ 3.32%] locator_check_foreign_key
|└ [ 1.90%] heap_attrinfo_start_with_index
| |└ [ 0.83%] heap_classrepr_get
| | |└ [ 11.07%] __GI___pthread_mutex_lock
| | |└ [ 5.34%] __lll_lock_wait
| | |└ [ 7.75%] __pthread_mutex_unlock_usercnt
| | |└ [ 4.29%] __lll_unlock_wake
| | |└ [ 0.15%] __GI___pthread_mutex_trylock
| |└ [ 0.29%] heap_attrinfo_recache_attrepr
| | |└ [ 0.11%] db_value_domain_init
| |└ [ 0.16%] hl_lea_alloc
|└ [ 3.09%] heap_attrinfo_end
| |└ [ 2.41%] heap_classrepr_free
| | |└ [ 11.07%] __GI___pthread_mutex_lock
| | |└ [ 5.34%] __lll_lock_wait
| | |└ [ 7.75%] __pthread_mutex_unlock_usercnt
| | |└ [ 4.29%] __lll_unlock_wake
| |└ [ 0.19%] heap_attrinfo_clear_dbvalues
| |└ [ 0.13%] mspace_free
|└ [ 1.79%] heap_attrinfo_read_dbvalues
| |└ [ 0.93%] heap_attrvalue_read
| | |└ [ 0.54%] heap_attrvalue_point_fixed
| | |└ [ 0.18%] tp_domain_disk_size
| | |└ [ 0.18%] heap_attrvalue_transform_to_dbvalue
| |└ [ 0.34%] or_rep_id
| | |└ [ 0.35%] or_header_size
| | |└ [ 0.18%] or_header_size@plt
| |└ [ 0.11%] pr_type_from_id
| |└ [ 0.10%] heap_attrinfo_recache
|└ [ 0.12%] heap_attrinfo_read_dbvalues@plt
└ [ 0.17%] heap_create_insert_context

```

orphan (root 경로에 등장하나 트리 도달 불가, >= 0.1%):

```

- [ 0.16%] lf_tran_start
- [ 0.13%] spage_get_record_data
- [ 0.13%] cubpl::session::is_thread_involved
- [ 0.13%] pthread_mutex_lock@plt
- [ 0.11%] pgbuf_get_holder
- [ 0.11%] prm_get_integer_value
- [ 0.10%] perfmon_add_stat
- [ 0.10%] perfmon_inc_stat

```

전체 함수 inclusive 표 (529개 전부, 컷오프 없음)

```

# locator_insert_force 경로 inclusive 함수 표 - 전체 529개 (컷오프 없음)
# denom = root_samples = 43840

```

%of_root	samples	func
100.000	43840	locator_insert_force
86.152	37769	heap_insert_logical
34.375	15070	heap_get_insert_location_with_lock
33.830	14831	heap_stats_find_best_page
27.753	12167	log_append_undoredo_crumbs
23.127	10139	heap_vpid_alloc
20.901	9163	lock_object
19.441	8523	lock_internal_perform_lock_object
18.150	7957	file_alloc
17.671	7747	heap_log_insert_physical
15.039	6593	prior_lsa_next_record_internal
14.865	6517	__GI__libc_malloc
13.955	6118	prior_lsa_alloc_and_copy_crumbs
12.331	5406	cubthread::lockfree_hashmap<lk_res_key, lk_res>::find_or_insert
12.256	5373	_int_malloc
11.955	5241	lf_hash_insert_internal
11.651	5108	lf_list_insert_internal
11.547	5062	cub_alloc
11.065	4851	__GI__pthread_mutex_lock
10.698	4690	spage_insert_at
9.968	4370	locator_add_or_remove_index_internal
9.856	4321	prior_lsa_gen_undoredo_record_from_crumbs
9.683	4245	__memmove_evex_unaligned_erms
9.366	4106	sysmalloc
9.268	4063	log_append_undoredo_data
9.250	4055	spage_insert_data
8.499	3726	lf_freelist_claim
8.250	3617	__GI__mprotect
7.753	3399	__pthread_mutex_unlock_usercnt
6.763	2965	lf_freelist_alloc_block
6.460	2832	lock_res_key_compare
5.839	2560	pgbuf_fix_release
5.342	2342	__lll_lock_wait
5.141	2254	pgbuf_claim_bcb_for_fix
4.601	2017	heap_stats_find_page_in_best_space
4.457	1954	LSA_SET_NULL
4.416	1936	log_zip
4.393	1926	fileio_init_lsa_of_page
4.377	1919	pgbuf_unfix
4.293	1882	__lll_unlock_wake
4.161	1824	file_perm_alloc
3.848	1687	log_sysop_commit_internal
3.501	1535	LZ4_resetStream_fast
3.317	1454	locator_check_foreign_key
3.301	1447	LOG_FIND_TDES
3.294	1444	log_append_sysop_end
3.264	1431	heap_get_class_name_alloc_if_diff
3.191	1399	__memset_evex_unaligned_erms
3.091	1355	lock_alloc_resource
3.086	1353	heap_attrinfo_end
2.958	1297	heap_vpid_init_new
2.701	1184	pgbuf_unlatch_bcb_upon_unfix
2.406	1055	heap_classrepr_free
2.130	934	log_sysop_end_logical_undo
2.007	880	spage_get_total_saved_spaces

2.005	879	log_append_undoredo_data2
1.930	846	spage_get_saved_spaces
1.909	837	heap_stats_add_bestspace
1.898	832	heap_attrinfo_start_with_index
1.793	786	heap_attrinfo_read_dbvalues
1.736	761	log_sysop_commit
1.512	663	spage_has_enough_total_space
1.494	655	prior_lsa_alloc_and_copy_data
1.476	647	lf_hash_find
1.432	628	spage_find_empty_slot_at
1.225	537	prior_lsa_copy_redo_data_to_node
1.147	503	heap_sancache_end
1.143	501	heap_sancache_quick_end
1.141	500	heap_mvcc_log_insert
1.131	496	spage_check_space
0.999	438	spage_max_space_for_new_record
0.931	408	heap_attrvalue_read
0.871	382	lf_list_find
0.833	365	lf_stack_pop
0.833	365	heap_classrepr_get
0.798	350	heap_get_class_record
0.778	341	logtb_get_current_mvccid
0.762	334	spage_insert
0.687	301	lock_event_set_xasl_id_to_entry
0.650	285	spage_find_empty_slot
0.648	284	cub_strdup
0.643	282	lock_find_class_entry
0.623	273	mht_get
0.614	269	lock_get_class_lock
0.604	265	pgbuf_unlatch_void_zone_bcb
0.595	261	log_sysop_start_atomic
0.573	251	mht_get2
0.538	236	heap_attrvalue_point_fixed
0.525	230	logtb_find_client_type
0.511	224	prior_lsa_copy_redo_crumbs_to_node
0.511	224	mvcc_is_mvcc_disabled_class
0.497	218	pgbuf_set_dirty_buffer_ptr
0.479	210	pgbuf_set_dirty
0.472	207	heap_attrvalue_get_key
0.470	206	prior_lsa_gen_record
0.452	198	prior_lsa_start_append
0.427	187	pgbuf_allocate_bcb
0.411	180	lock_initialize_entry_as_granted
0.408	179	prior_lsa_copy_undo_crumbs_to_node
0.406	178	pgbuf_lru_add_new_bcb_to_top
0.395	173	spage_get_record
0.392	172	pgbuf_set_lsa
0.390	171	logtb_find_tranid
0.381	167	pgbuf_unlatch_thrd_holder
0.376	165	xlogtb_reset_wait_msecs
0.367	161	vacuum_get_log_blockid
0.354	155	or_header_size
0.349	153	log_diff
0.344	151	or_rep_id
0.344	151	pgbuf_get_bcb_from_invalid_list
0.338	148	pgbuf_find_thrd_holder
0.335	147	__gthread_mutex_unlock

```

0.326      143 prior_lsa_copy_undo_data_to_node
0.324      142 ATOMIC_INC_32<int, int>
0.324      142 pgbuf_lru_adjust_zones
0.317      139 log_sysop_start
0.315      138 spage_find_slot
0.306      134 btree_insert
0.303      133 heap_page_get_vacuum_status
0.301      132 heap_clean_get_context
0.299      131 unlink_chunk.isra.2
0.297      130 lock_insert_into_tran_hold_list
0.297      130 LOG_FIND_CURRENT_TDES
0.294      129 cubpl::get_session
0.285      125 heap_attrinfo_recache_attrepr
0.283      124 log_tdes::unlock_topop
0.274      120 pgbuf_is_lsa_temporary
0.271      119 heap_sancache_quick_start_root_hfid
0.265      116 heap_get_last_version
0.260      114 __tls_get_addr
0.253      111 pgbuf_wakeup_reader_writer
0.251      110 set_waiter_exists
0.249      109 prior_lsa_append_data
0.246      108 log_prior_lsa_append_advance_when_doesnot_fit
0.228      100 __strlen_evex
0.228      100 pgbuf_ordered_unfix
0.217      95  xdisk_get_purpose
0.210      92  pgbuf_bcb_change_zone
0.210      92  pgbuf_replace_watcher
0.208      91  pgbuf_bcb_register_hit_for_lru
0.205      90  session_get_pl_session
0.203      89  heap_attrinfo_access
0.203      89  log_tdes::lock_topop
0.196      86  ATOMIC_CAS_32<int, int, int>
0.194      85  pgbuf_remove_thrd_holder
0.194      85  heap_attrinfo_clear_dbvalues
0.194      85  lf_callback_vpid_hash
0.192      84  ATOMIC_CAS_ADDR<void>
0.189      83  heap_page_update_chain_after_mvcc_op
0.185      81  thread_get_thread_entry_info
0.182      80  or_header_size@plt
0.180      79  tp_domain_disk_size
0.178      78  mht_put_internal
0.176      77  heap_attrvalue_transform_to_dbvalue
0.173      76  heap_create_insert_context
0.173      76  mht_rem
0.171      75  perfmon_is_perf_tracking
0.169      74  pgbuf_ordered_set_dirty_and_free
0.167      73  er_errid
0.160      70  lf_tran_start
0.155      68  hl_lea_alloc
0.151      66  pgbuf_remove_watcher
0.148      65  __GI__pthread_mutex_trylock
0.148      65  logtb_find_wait_msecs
0.146      64  spage_initialize
0.144      63  cubthread::get_entry
0.139      61  log_tdes::is_system_worker_transaction
0.139      61  spage_add_new_slot
0.137      60  mvcc_is_mvcc_disabled_class@plt

```

```

0.135      59  cubpl::session::is_sp_running
0.135      59  spage_get_record_data
0.135      59  log_zip@plt
0.132      58  mspace_free
0.132      58  cubpl::session::is_thread_involved
0.130      57  logpb_get_memsize
0.128      56  boot_find_root_heap
0.128      56  pgbuf_lru_adjust_zone1
0.125      55  pgbuf_lru_fall_bcb_to_zone_3
0.125      55  pgbuf_search_hash_chain
0.125      55  pthread_mutex_lock@plt
0.123      54  or_class_name
0.123      54  log_prior_lsa_append_align
0.121      53  heap_attrinfo_read_dbvalues@plt
0.119      52  pgbuf_get_vpid_ptr
0.119      52  log_can_skip_undo_logging
0.114      50  btree_mvcc_info_from_heap_mvcc_header
0.112      49  lock_escalate_if_needed
0.112      49  pgbuf_get_holder
0.112      49  heap_insert_adjust_recdes_header
0.112      49  pr_type_from_id
0.109      48  db_value_domain_init
0.107      47  prm_get_integer_value
0.105      46  heap_attrinfo_recache
0.105      46  log_sysop_end_final
0.103      45  spage_get_record@plt
0.100      44  logtb_get_current_tran_index
0.100      44  pgbuf_lock_page
0.100      44  perfmon_add_stat
0.100      44  perfmon_inc_stat
0.098      43  heap_hash_vpid
0.096      42  heap_hash_hfid
0.096      42  lf_tran_end@plt
0.096      42  LZ4_compress_fast_extState@plt
0.094      41  log_zip_realloc_if_needed
0.094      41  LZ4_compress_fast_extState
0.094      41  or_class_name@plt
0.091      40  pgbuf_bcb_register_fix
0.091      40  spage_find_free_slot
0.089      39  pgbuf_unfix@plt
0.089      39  mht_rem2
0.087      38  LSA_COPY
0.087      38  btree_set_mvcc_header_ids_for_update
0.087      38  or_chn
0.087      38  spage_verify_header
0.084      37  cuberr::context::get_thread_local_error
0.082      36  log_sysop_get_tran_index_and_tdes
0.080      35  hl_lea_free
0.080      35  btree_insert@plt
0.080      35  pgbuf_latch_bcb_upon_fix
0.080      35  heap_compare_hfid
0.080      35  malloc@plt
0.080      35  logtb_get_current_mvccid@plt
0.080      35  logtb_is_active
0.078      34  malloc Consolidate
0.075      33  log_zip_realloc_if_needed@plt
0.075      33  mr_data_readval_int

```



```

0.075      33 mspace_malloc
0.075      33 lock_res_key_hash
0.075      33 __GI___pthread_mutex_unlock
0.075      33 heap_page_get_vacuum_status@plt
0.075      33 heap_sancache_quick_start_internal
0.073      32 btree_set_mvcc_header_ids_for_update@plt
0.071      31 pgbuf_notify_vacuum_follows
0.071      31 heap_prepare_get_context@plt
0.071      31 __gthread_mutex_lock
0.071      31 heap_is_big_length
0.071      31 pgbuf_is_bcb_victimizable
0.068      30 pgbuf_bcb_set_dirty
0.068      30 pgbuf_get_volume_id
0.068      30 cuberr::context::get_thread_local_context
0.068      30 pgbuf_is_temp_lsa
0.068      30 pgbuf_get_vpid_ptr@plt
0.066      29 __GI___libc_free
0.066      29 heap_get_class_name_alloc_if_diff@plt
0.066      29 memcpy@plt
0.066      29 pgbuf_is_temporary_volume
0.064      28 prior_lsa_end_append
0.064      28 pr_clear_value
0.062      27 heap_clean_get_context@plt
0.062      27 LZ4_compressBound@plt
0.062      27 heap_attrinfo_start_with_index@plt
0.062      27 oid_check_cached_class_oid
0.062      27 logtb_find_tranid@plt
0.062      27 heap_sancache_check_with_hfid
0.059      26 pthread_mutex_unlock@plt
0.059      26 cubthread::lockfree_hashmap<lk_res_key, lk_res>::find_or_insert@plt
0.059      26 heap_sancache_quick_start_root_hfid@plt
0.057      25 logtb_find_current_tranid@plt
0.057      25 xdisk_get_purpose@plt
0.057      25 or_chn@plt
0.057      25 heap_attrvalue_locate
0.057      25 free@plt
0.055      24 heap_init_get_context
0.055      24 log_tdes::is_system_transaction@plt
0.055      24 pgbuf_get_tde_algorithm@plt
0.052      23 pgbuf_set_dirty@plt
0.052      23 pgbuf_find_current_wait_msecs
0.052      23 boot_find_root_heap@plt
0.052      23 pgbuf_ordered_unfix@plt
0.052      23 heap_scan_pb_lock_and_fetch
0.052      23 pgbuf_replace_watcher@plt
0.050      22 heap_get_record_data_when_all_ready@plt
0.050      22 heap_classrepr_get@plt
0.050      22 lock_object@plt
0.050      22 heap_attrvalue_locate@plt
0.050      22 __tls_get_addr@plt
0.050      22 prior_lsa_alloc_and_copy_crumbs@plt
0.048      21 or_rep_id@plt
0.048      21 pgbuf_bcb_update_flags
0.048      21 log_tdes::is_system_worker_transaction@plt
0.048      21 LSA_ISNULL
0.046      20 log_sysop_end_begin
0.046      20 pgbuf_get_volume_id@plt

```

```

0.046      20 log_append_undoredo_crumbs@plt
0.046      20 spage_max_record_size
0.046      20 spage_set_slot
0.043      19 LZ4_compressBound
0.043      19 LOG_PRIOR_LSA_LAST_APPEND_OFFSET@plt
0.043      19 log_is_in_crash_recovery
0.043      19 __bswap_32
0.043      19 oid_check_cached_class_oid@plt
0.043      19 heap_get_class_record@plt
0.043      19 spage_is_unknown_slot
0.043      19 pgbuf_get_page_id
0.041      18 pgbuf_lru_add_bcb_to_top
0.041      18 session_get_session_state
0.039      17 fileio_set_page_lsa
0.039      17 db_private_alloc_release
0.039      17 pgbuf_ordered_fix_release
0.039      17 pgbuf_lockfree_unfix_ro
0.039      17 heap_attrinfo_end@plt
0.036      16 __l1_unlock_wake_private
0.036      16 prn_get_bool_value
0.036      16 spage_find_free_slot@plt
0.036      16 lf_tran_end
0.036      16 lf_tran_start@plt
0.036      16 lock_get_hash_value
0.036      16 prior_lsa_next_record@plt
0.034      15 lock_get_class_lock@plt
0.034      15 pgbuf_set_bcb_page_vpid
0.032      14 log_is_in_crash_recovery@plt
0.032      14 lock_check_escalate
0.032      14 heap_attrvalue_get_key@plt
0.032      14 log_append_get_zip_redo@plt
0.032      14 lock_get_new_entry
0.032      14 vacuum_is_mvccid_vacuumed
0.032      14 pgbuf_fix_release@plt
0.032      14 pgbuf_ordered_fix_release@plt
0.032      14 pgbuf_ordered_set_dirty_and_free@plt
0.030      13 cub_free
0.030      13 heap_insert_logical@plt
0.030      13 tp_domain_disk_size@plt
0.030      13 db_private_free_release
0.030      13 _int_free
0.030      13 log_prior_lsa_append_add_align
0.030      13 db_private_alloc_release@plt
0.027      12 pgbuf_set_lsa@plt
0.027      12 pgbuf_get_tde_algorithm
0.027      12 cuberr::context::get_thread_local_error@plt
0.027      12 pr_clear_value@plt
0.027      12 cubthread::get_entry@plt
0.027      12 heap_get_class_name@plt
0.027      12 file_table_add_full_sector
0.027      12 heap_sancache::start_area@plt
0.027      12 pr_type_from_id@plt
0.027      12 spage_find_slot_for_insert
0.025      11 rmutex_unlock
0.025      11 file_alloc@plt
0.025      11 or_get_int
0.025      11 spage_insert_at@plt

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0.025      11 get_impl
0.025      11 spage_max_record_size@plt
0.025      11 log_tdes::is_system_transaction
0.025      11 btree_mvcc_info_from_heap_mvcc_header@plt
0.025      11 strlen@plt
0.025      11 pgbuf_is_lsa_temporary@plt
0.025      11 heap_sancache::start_area
0.023      10 rmutex_lock
0.023      10 heap_attrinfo_clear_dbvalues@plt
0.023      10 log_data_addr::log_data_addr@plt
0.023      10 log_append_undoredo_data@plt
0.023      10 heap_get_class_name
0.023      10 file_partsect_alloc
0.023      10 LSA_LE
0.023      10 pgbuf_set_page_ptype
0.023      10 log_append_get_zip_redo
0.023      10 heap_sancache::end_area
0.023      10 file_perm_expand
0.023      10 vacuum_get_log_blockid@plt
0.023      10 ATOMIC_INC_64<long, int>
0.023      10 oid_is_serial
0.021       9 logtb_get_current_tran_index@plt
0.021       9 file_log_fhead_alloc
0.021       9 bit64_count_trailing_zeros@plt
0.021       9 heap_get_last_page
0.021       9 check_supplemental_log
0.021       9 cubpl::get_session@plt
0.021       9 logtb_find_client_type@plt
0.021       9 pgbuf_set_tde_algorithm
0.021       9 heap_init_get_context@plt
0.021       9 pgbuf_get_page_id@plt
0.021       9 logtb_find_wait_msecs@plt
0.021       9 heap_classrepr_entry_remove_from_LRU
0.021       9 log_skip_logging@plt
0.018       8 heap_sancache_end@plt
0.018       8 hl_lea_free@plt
0.018       8 cub_free@plt
0.018       8 heap_compare_vpid
0.018       8 locator_add_or_remove_index
0.018       8 mmon_is_memory_monitor_enabled
0.018       8 or_init
0.018       8 logtb_is_active@plt
0.018       8 xlogtb_reset_wait_msecs@plt
0.018       8 spage_initialize@plt
0.018       8 spage_number_of_slots
0.018       8 log_append_undoredo_data2@plt
0.018       8 heap_classrepr_free@plt
0.016       7 pgbuf_lru_boost_bcb
0.016       7 prior_lsa_next_record
0.016       7 mht_rem@plt
0.016       7 heap_get_mvcc_header@plt
0.016       7 LOG_PRIOR_LSA_LAST_APPEND_OFFSET
0.016       7 cuberr::context::get_current_error_level
0.016       7 __GI___pthread_mutex_init
0.016       7 cuberr::context::get_thread_local_context@plt
0.016       7 heap_attrinfo_access@plt
0.016       7 bit64_count_trailing_ones

```

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0.016      7 pgbuf_unlock_page
0.016      7 lf_freelist_transport
0.014      6 cubthread::worker_pool::core::worker::run
0.014      6 css_server_task::execute
0.014      6 slocator_repl_force
0.014      6 execute_native_thread_routine
0.014      6 start_thread
0.014      6 cubthread::worker_pool::core::worker::execute_current_task
0.014      6 __GI___clone
0.014      6 net_server_request
0.014      6 pgbuf_set_tde_algorithm@plt
0.014      6 pgbuf_bcb_check_and_reset_fix_and_avoid_dealloc
0.014      6 log_tdes::on_sysop_start
0.014      6 mht_adjust_lru_list@plt
0.014      6 logtb_find_current_tranid
0.014      6 mht_put_new@plt
0.014      6 log_diff@plt
0.014      6 lf_hash_find@plt
0.014      6 btree_insert_internal
0.014      6 lf_stack_pop@plt
0.014      6 mspace_malloc@plt
0.014      6 pgbuf_allocate_thrd_holder_entry
0.014      6 mht_rem2@plt
0.011      5 cubthread::is_single_thread@plt
0.011      5 oid_is_serial@plt
0.011      5 pgbuf_get_lsa@plt
0.011      5 pgbuf_set_page_ptype@plt
0.011      5 logtb_is_interrupted
0.011      5 log_append_get_zip_undo@plt
0.011      5 heap_sancache::end_area@plt
0.011      5 lock_alloc_entry
0.011      5 log_sysop_end_logical_undo@plt
0.011      5 perfmon_is_perf_tracking_and_active
0.011      5 PGBUF_THREAD_HAS_PRIVATE_LRU
0.011      5 vacuum_is_mvccid_vacuued@plt
0.011      5 PRIM_SET_NULL
0.011      5 log_append_get_zip_undo
0.011      5 mht_adjust_lru_list
0.011      5 log_sysop_start@plt
0.011      5 __lll_lock_wait_private
0.009      4 bit64_count_trailing_ones@plt
0.009      4 log_tdes::on_sysop_end
0.009      4 lf_list_find@plt
0.009      4 lf_hash_find_or_insert
0.009      4 pgbuf_notify_vacuum_follows@plt
0.009      4 mht_get@plt
0.009      4 bit64_set
0.009      4 prior_lsa_alloc_and_copy_data@plt
0.009      4 prior_update_header_mvcc_info
0.009      4 bit64_count_trailing_zeros
0.009      4 pgbuf_hash_func_mirror
0.009      4 er_errid@plt
0.009      4 memset@plt
0.009      4 logpb_get_memsize@plt
0.009      4 pgbuf_add_watch_instance_internal
0.009      4 sys_alloc
0.007      3 cub_realloc

```

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0.007      3 log_append_realloc_data_ptr
0.007      3 xlocator_repl_force
0.007      3 lf_hash_find_or_insert@plt
0.007      3 hl_lea_alloc@plt
0.007      3 db_value_domain_init@plt
0.007      3 heap_get_last_version@plt
0.007      3 rmutex_unlock@plt
0.007      3 heap_is_big_length@plt
0.007      3 thread_get_tran_entry
0.007      3 mspace_free@plt
0.007      3 heap_classrepr_get_from_record
0.007      3 log_tdes::is_active_worker_transaction@plt
0.007      3 heap_link_watchers
0.007      3 db_private_get_heapid_from_thread
0.007      3 spage_number_of_slots@plt
0.007      3 pgbuf_bcb_is_async_flush_request
0.007      3 heap_insert_physical
0.007      3 pgbuf_should_move_private_to_shared
0.007      3 lf_freelist_claim@plt
0.007      3 spage_check_record_for_insert
0.007      3 db_make_null
0.007      3 log_sysop_commit@plt
0.007      3 db_value_domain_type
0.007      3 bit64_set@plt
0.005      2 heap_create_insert_context@plt
0.005      2 heap_sancache_end_internal
0.005      2 spage_max_space_for_new_record@plt
0.005      2 cubpl::session::is_sp_running@plt
0.005      2 pthread_mutex_init@plt
0.005      2 log_tdes::is_active_worker_transaction
0.005      2 lock_res_key_copy
0.005      2 pthread_mutex_trylock@plt
0.005      2 db_private_free_release@plt
0.005      2 lock_create_search_key
0.005      2 cuberr::context::get_current_error_level@plt
0.005      2 cubthread::entry::get_id@plt
0.005      2 lock_conv
0.005      2 log_tdes::lock_topop@plt
0.005      2 logtb_get_check_interrupt
0.005      2 vacuum_produce_log_block_data
0.005      2 heap_prepare_get_context
0.002      1 mvcctable::get_global_oldest_visible
0.002      1 pgbuf_get_lsa
0.002      1 LZ4_resetStream_fast@plt
0.002      1 __GI___qsort_r
0.002      1 file_extdata_find_not_full
0.002      1 db_User_page_size
0.002      1 lock_init_entry
0.002      1 heap_stats_entry_free
0.002      1 log_skip_logging
0.002      1 mht_get2@plt
0.002      1 log_tdes::unlock_topop@plt
0.002      1 log_sysop_start_atomic@plt
0.002      1 heap_get_mvcc_header
0.002      1 logtb_is_interrupted@plt
0.002      1 file_partsect_is_empty
0.002      1 disk_compare_vsids

```

```

0.002      1 file_extdata_find_ordered
0.002      1 util_bsearch
0.002      1 msort_with_tmp.part.0
0.002      1 session_get_pl_session@plt
0.002      1 file_partsect_set_bit
0.002      1 log_data_addr::log_data_addr
0.002      1 mht_put_new
0.002      1 pgbuf_lockfree_fix_ro
0.002      1 vacuum_is_thread_vacuum_worker
0.002      1 lock_dealloc_entry
0.002      1 pgbuf_assign_direct_victim
0.002      1 heap_get_record_data_when_all_ready
0.002      1 log_tdes::on_sysop_end@plt
0.002      1 lf_freelist_transport@plt
0.002      1 heap_unfix_watchers
0.002      1 cubpl::session::is_thread_involved@plt
0.002      1 pgbuf_check_bcb_page_vpid
0.002      1 lock_compat

```

전체 콜체인 (모든 thread·서버 기능 포함, % of total)

위 §콜트리는 locator_insert_force subtree(=100%)만 본 것이고, 아래는 캡처된 전체 **sample**을 thread(comm)별로 묶은 콜체인입니다. 노드 % = 전체 **sample(147,236)** 대비. stack truncate 를 감안해 각 sample 의 최외곽 프레임을 thread 노드에 매달아 100% 를 덮습니다.

원본: analysis/calltree_full.txt, 전 함수(1,840개) 평탄 표: analysis/inclusive_total.txt, 전체 flamegraph: analysis/flame_full.svg.

thread(서버 스레드) 분포

% of total	samples	thread	역할
69.65	102,544	transaction	워커(insert 실행)
13.16	19,374	connections	클라이언트 연결 처리
5.33	7,847	vacuum	vacuum 워커
3.82	5,618	dwb-flush-block	double write buffer flush
3.72	5,483	log-flush	WAL flush
1.69	2,488	coordinator	트랜잭션 coordinator
0.85	1,253	vacuum-master	vacuum 마스터
0.73	1,080	dwb-file-sync	dwb 파일 sync
1.05	1,549	기타 9종	pgbuf-flush/maintain, deadlock-detect, log-clock 등

전체 콜트리 (노드컷 0.2% of total, thread컷 0.3%, MIN_EDGE 30)

```

# 전체 on-CPU 콜체인 (모든 thread·함수, 노드 % = of total_samples=147236)
# 함수 노드 컷 0.2% / thread 컷 0.3% / MIN_EDGE 30

```

```

[100.00%] (ALL on-CPU)  total_samples=147236
├─ [ 69.65%] «transaction»  (102544 smp)
│   └─ [ 29.78%] locator_insert_force
│       └─ [ 25.66%] heap_insert_logical
│           └─ [ 10.24%] heap_get_insert_location_with_lock
│               └─ [ 11.94%] heap_stats_find_best_page

```

```

└─ [ 6.89%] heap_vpid_alloc
└─ [ 5.46%] file_alloc
└─ [ 3.91%] pgbuf_fix_release
└─ [ 1.56%] pgbuf_claim_bcb_for_fix
└─ [ 1.33%] fileio_init_lsa_of_page
└─ [ 1.35%] LSA_SET_NULL
└─ [ 0.39%] pgbuf_search_hash_chain
└─ [ 0.40%] __GI__pthread_mutex_trylock
└─ [ 0.30%] pgbuf_latch_bcb_upon_fix
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __lll_unlock_wake
└─ [ 1.25%] file_perm_alloc
└─ [ 0.62%] log_append_undoredo_data2
└─ [ 19.52%] log_append_undoredo_crumbs
└─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
└─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
└─ [ 9.96%] log_zip
└─ [ 8.84%] LZ4_compress_fast_extState
└─ [ 1.72%] LZ4_read_ARCH
└─ [ 0.78%] LZ4_read32
└─ [ 0.47%] LZ4_initStream
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.21%] LZ4_writeLE16
└─ [ 1.06%] LZ4_resetStream_fast
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI__libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI__mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI__libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
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└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 0.24%] LOG_FIND_CURRENT_TDES
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI__libc_malloc
└─ [ 6.52%] _int_malloc

```

```

└ [ 4.20%] sysmalloc
  └ [ 3.59%] __GI___mprotect
└ [ 0.68%] malloc_consolidate
  └ [ 0.32%] unlink_chunk.isra.2
    └ [ 0.32%] unlink_chunk.isra.2
└ [ 9.96%] log_zip
  └ [ 8.84%] LZ4_compress_fast_extState
    └ [ 1.72%] LZ4_read_ARCH
      └ [ 0.78%] LZ4_read32
        └ [ 0.47%] LZ4_initStream
          └ [ 1.36%] __memset_evex_unaligned_erms
            └ [ 0.21%] LZ4_writeLE16
              └ [ 1.06%] LZ4_resetStream_fast
                └ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.59%] prior_lsa_copy_undo_data_to_node
  └ [ 6.08%] cub_alloc
    └ [ 7.24%] __GI___libc_malloc
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└ [ 0.40%] prior_lsa_copy_redo_data_to_node
  └ [ 6.08%] cub_alloc
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              └ [ 0.32%] unlink_chunk.isra.2
                └ [ 0.32%] unlink_chunk.isra.2
          └ [ 3.69%] __memmove_evex_unaligned_erms
            └ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 7.94%] prior_lsa_next_record_internal
  └ [ 5.61%] __GI___pthread_mutex_lock
    └ [ 2.75%] _lll_lock_wait
      └ [ 4.27%] __pthread_mutex_unlock_usercnt
        └ [ 2.37%] _lll_unlock_wake
          └ [ 0.22%] prior_lsa_start_append
            └ [ 0.22%] vacuum_get_log_blockid
              └ [ 1.90%] LOG_FIND_TDES
└ [ 7.88%] log_append_undoredo_data
  └ [ 19.52%] log_append_undoredo_crumbs
    └ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
      └ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
        └ [ 9.96%] log_zip
          └ [ 8.84%] LZ4_compress_fast_extState
            └ [ 1.72%] LZ4_read_ARCH
              └ [ 0.78%] LZ4_read32
                └ [ 0.47%] LZ4_initStream
                  └ [ 1.36%] __memset_evex_unaligned_erms
                    └ [ 0.21%] LZ4_writeLE16
                      └ [ 1.06%] LZ4_resetStream_fast
                        └ [ 1.36%] __memset_evex_unaligned_erms
└ [ 6.08%] cub_alloc
  └ [ 7.24%] __GI___libc_malloc

```



```

└─ [ 6.52%] _int_malloc
    └─ [ 4.20%] sysmalloc
        └─ [ 3.59%] __GI___mprotect
            └─ [ 0.68%] malloc_consolidate
                └─ [ 0.32%] unlink_chunk.isra.2
                    └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
                                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
                                └─ [ 0.32%] unlink_chunk.isra.2
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                        └─ [ 0.32%] unlink_chunk.isra.2
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        └─ [ 1.72%] LZ4_read_ARCH
            └─ [ 0.78%] LZ4_read32
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                        └─ [ 0.21%] LZ4_writeLE16
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    └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
    └─ [ 6.08%] cub_alloc
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            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
                                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.40%] prior_lsa_copy_redo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc

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```

└ [ 4.20%] sysmalloc
  └ [ 3.59%] __GI___mprotect
└ [ 0.68%] malloc_consolidate
  └ [ 0.32%] unlink_chunk.isra.2
    └ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
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└ [ 7.94%] prior_lsa_next_record_internal
  └ [ 5.61%] __GI___pthread_mutex_lock
    └ [ 2.75%] __lll_lock_wait
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  └ [ 0.22%] prior_lsa_start_append
    └ [ 0.22%] vacuum_get_log_blockid
└ [ 1.90%] LOG_FIND_TDES
└ [ 0.88%] heap_vpid_init_new
└ [ 7.88%] log_append_undoredo_data
  └ [ 19.52%] log_append_undoredo_crumbs
    └ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
      └ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
        └ [ 9.96%] log_zip
          └ [ 8.84%] LZ4_compress_fast_extState
            └ [ 1.72%] LZ4_read_ARCH
              └ [ 0.78%] LZ4_read32
                └ [ 0.47%] LZ4_initStream
                  └ [ 1.36%] __memset_evex_unaligned_erms
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└ [ 6.08%] cub_alloc
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        └ [ 4.20%] sysmalloc
          └ [ 3.59%] __GI___mprotect
            └ [ 0.68%] malloc_consolidate
              └ [ 0.32%] unlink_chunk.isra.2
                └ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 0.24%] LOG_FIND_CURRENT_TDES

```

```

└ [ 1.90%] LOG_FIND_TDES
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 9.96%] log_zip
└ [ 8.84%] LZ4_compress_fast_extState
└ [ 1.72%] LZ4_read_ARCH
└ [ 0.78%] LZ4_read32
└ [ 0.47%] LZ4_initStream
└ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.21%] LZ4_writeLE16
└ [ 1.06%] LZ4_resetStream_fast
└ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.59%] prior_lsa_copy_undo_data_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.40%] prior_lsa_copy_redo_data_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 7.94%] prior_lsa_next_record_internal
└ [ 5.61%] __GI__pthread_mutex_lock
└ [ 2.75%] __lll_lock_wait
└ [ 4.27%] __pthread_mutex_unlock_usercnt
└ [ 2.37%] __lll_unlock_wake
└ [ 0.22%] prior_lsa_start_append
└ [ 0.22%] vacuum_get_log_blockid
└ [ 1.90%] LOG_FIND_TDES
└ [ 0.23%] spage_insert
└ [ 0.21%] pgbuf_set_dirty
└ [ 0.64%] log_sysop_end_logical_undo
└ [ 1.17%] log_sysop_commit_internal
└ [ 0.99%] log_append_sysop_end
└ [ 7.94%] prior_lsa_next_record_internal
└ [ 5.61%] __GI__pthread_mutex_lock
└ [ 2.75%] __lll_lock_wait
└ [ 4.27%] __pthread_mutex_unlock_usercnt
└ [ 2.37%] __lll_unlock_wake
└ [ 0.22%] prior_lsa_start_append

```

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└─ [ 0.22%] vacuum_get_log_blockid
└─ [ 0.45%] prior_lsa_alloc_and_copy_data
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.39%] log_tdes::unlock_topop
└─ [ 0.38%] cubpl::get_session
└─ [ 0.24%] session_get_pl_session
└─ [ 0.27%] thread_get_thread_entry_info
└─ [ 0.21%] cubthread::get_entry
└─ [ 7.94%] prior_lsa_next_record_internal
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __lll_lock_wait
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __lll_unlock_wake
└─ [ 0.22%] prior_lsa_start_append
└─ [ 0.22%] vacuum_get_log_blockid
└─ [ 1.99%] pgbuf_unfix
└─ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __lll_unlock_wake
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __lll_lock_wait
└─ [ 7.88%] log_append_undoredo_data
└─ [ 19.52%] log_append_undoredo_crumbs
└─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
└─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
└─ [ 9.96%] log_zip
└─ [ 8.84%] LZ4_compress_fast_extState
└─ [ 1.72%] LZ4_read_ARCH
└─ [ 0.78%] LZ4_read32
└─ [ 0.47%] LZ4_initStream
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.21%] LZ4_writeLE16
└─ [ 1.06%] LZ4_resetStream_fast
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2

```

```

└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 0.24%] LOG_FIND_CURRENT_TDES
└ [ 1.90%] LOG_FIND_TDES
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 9.96%] log_zip
└ [ 8.84%] LZ4_compress_fast_extState
└ [ 1.72%] LZ4_read_ARCH
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└ [ 0.47%] LZ4_initStream
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└ [ 0.21%] LZ4_writeLE16
└ [ 1.06%] LZ4_resetStream_fast
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└ [ 0.59%] prior_lsa_copy_undo_data_to_node
└ [ 6.08%] cub_alloc
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└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.40%] prior_lsa_copy_redo_data_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2

```

```

└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 7.94%] prior_lsa_next_record_internal
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.75%] __lll_lock_wait
└─ [ 2.37%] __lll_unlock_wake
└─ [ 0.22%] prior_lsa_start_append
└─ [ 0.22%] vacuum_get_log_blockid
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 0.52%] log_sysop_commit
└─ [ 1.17%] log_sysop_commit_internal
└─ [ 0.99%] log_append_sysop_end
└─ [ 7.94%] prior_lsa_next_record_internal
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.75%] __lll_lock_wait
└─ [ 2.37%] __lll_unlock_wake
└─ [ 0.22%] prior_lsa_start_append
└─ [ 0.22%] vacuum_get_log_blockid
└─ [ 0.45%] prior_lsa_alloc_and_copy_data
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.39%] log_tdes::unlock_topop
└─ [ 0.38%] cubpl::get_session
└─ [ 0.24%] session_get_pl_session
└─ [ 0.27%] thread_get_thread_entry_info
└─ [ 0.21%] cubthread::get_entry
└─ [ 0.57%] heap_stats_add_bestspace
└─ [ 0.34%] mht_get
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __lll_lock_wait
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __lll_unlock_wake
└─ [ 1.99%] pgbuf_unfix
└─ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __lll_unlock_wake
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __lll_lock_wait
└─ [ 0.61%] log_sysop_start
└─ [ 0.22%] log_tdes::lock_topop

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└─ [ 0.38%] cubpl::get_session
    └─ [ 0.24%] session_get_pl_session
        └─ [ 0.27%] thread_get_thread_entry_info
            └─ [ 0.21%] cubthread::get_entry
└─ [ 0.69%] rmutex_lock
    └─ [ 5.61%] __GI__pthread_mutex_lock
        └─ [ 2.75%] __lll_lock_wait
            └─ [ 0.27%] thread_get_thread_entry_info
                └─ [ 0.21%] cubthread::get_entry
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 7.88%] log_append_undoredo_data
    └─ [ 19.52%] log_append_undoredo_crumbs
        └─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
            └─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
                └─ [ 9.96%] log_zip
                    └─ [ 8.84%] LZ4_compress_fast_extState
                        └─ [ 1.72%] LZ4_read_ARCH
                            └─ [ 0.78%] LZ4_read32
                                └─ [ 0.47%] LZ4_initStream
                                    └─ [ 1.36%] __memset_evex_unaligned_erms
                                        └─ [ 0.21%] LZ4_writeLE16
                                            └─ [ 1.06%] LZ4_resetStream_fast
                                                └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI__libc_malloc
        └─ [ 6.52%] _int_malloc
            └─ [ 4.20%] sysmalloc
                └─ [ 3.59%] __GI__mprotect
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
                            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI__libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI__mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
                                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI__libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI__mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
                                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 0.24%] LOG_FIND_CURRENT_TDES
    └─ [ 1.90%] LOG_FIND_TDES
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI__libc_malloc
        └─ [ 6.52%] _int_malloc
            └─ [ 4.20%] sysmalloc
                └─ [ 3.59%] __GI__mprotect

```

```

└─ [ 0.68%] malloc_consolidate
    └─ [ 0.32%] unlink_chunk.isra.2
        └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 9.96%] log_zip
    └─ [ 8.84%] LZ4_compress_fast_extState
        └─ [ 1.72%] LZ4_read_ARCH
        └─ [ 0.78%] LZ4_read32
        └─ [ 0.47%] LZ4_initStream
            └─ [ 1.36%] __memset_evex_unaligned_erms
            └─ [ 0.21%] LZ4_writeLE16
        └─ [ 1.06%] LZ4_resetStream_fast
            └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
                        └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.40%] prior_lsa_copy_redo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
                        └─ [ 0.32%] unlink_chunk.isra.2
        └─ [ 3.69%] __memmove_evex_unaligned_erms
    └─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 7.94%] prior_lsa_next_record_internal
    └─ [ 5.61%] __GI___pthread_mutex_lock
        └─ [ 2.75%] __lll_lock_wait
    └─ [ 4.27%] __pthread_mutex_unlock_usercnt
        └─ [ 2.37%] __lll_unlock_wake
    └─ [ 0.22%] prior_lsa_start_append
        └─ [ 0.22%] vacuum_get_log_blockid
    └─ [ 1.90%] LOG_FIND_TDES
└─ [ 1.37%] heap_stats_find_page_in_bestspace
    └─ [ 0.57%] heap_stats_add_bestspace
        └─ [ 0.34%] mht_get
        └─ [ 5.61%] __GI___pthread_mutex_lock
            └─ [ 2.75%] __lll_lock_wait
        └─ [ 4.27%] __pthread_mutex_unlock_usercnt
            └─ [ 2.37%] __lll_unlock_wake
    └─ [ 0.30%] spage_max_space_for_new_record
        └─ [ 0.60%] spage_get_total_saved_spaces
            └─ [ 0.58%] spage_get_saved_spaces
                └─ [ 0.45%] lf_hash_find
                    └─ [ 0.26%] lf_list_find
                └─ [ 0.47%] spage_has_enough_total_space
    └─ [ 5.61%] __GI___pthread_mutex_lock
        └─ [ 2.75%] __lll_lock_wait
    └─ [ 4.27%] __pthread_mutex_unlock_usercnt
        └─ [ 2.37%] __lll_unlock_wake

```


[illegible]

```

└─ [ 5.61%] __GI__pthread_mutex_lock
    └─ [ 2.75%] __lll_lock_wait
    └─ [ 4.27%] __pthread_mutex_unlock_usercnt
        └─ [ 2.37%] __lll_unlock_wake
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
    └─ [ 2.37%] __lll_unlock_wake
└─ [ 5.61%] __GI__pthread_mutex_lock
    └─ [ 2.75%] __lll_lock_wait
└─ [ 0.20%] lock_get_class_lock
    └─ [ 5.61%] __GI__pthread_mutex_lock
        └─ [ 2.75%] __lll_lock_wait
        └─ [ 4.27%] __pthread_mutex_unlock_usercnt
            └─ [ 2.37%] __lll_unlock_wake
└─ [ 0.20%] lock_find_class_entry
    └─ [ 5.61%] __GI__pthread_mutex_lock
        └─ [ 2.75%] __lll_lock_wait
        └─ [ 4.27%] __pthread_mutex_unlock_usercnt
            └─ [ 2.37%] __lll_unlock_wake
└─ [ 11.44%] heap_log_insert_physical
    └─ [ 19.52%] log_append_undoredo_crumbs
        └─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
            └─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
                └─ [ 9.96%] log_zip
                    └─ [ 8.84%] LZ4_compress_fast_extState
                        └─ [ 1.72%] LZ4_read_ARCH
                        └─ [ 0.78%] LZ4_read32
                        └─ [ 0.47%] LZ4_initStream
                            └─ [ 1.36%] __memset_evex_unaligned_erms
                                └─ [ 0.21%] LZ4_writeLE16
                                    └─ [ 1.06%] LZ4_resetStream_fast
                                        └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI__libc_malloc
        └─ [ 6.52%] _int_malloc
            └─ [ 4.20%] sysmalloc
                └─ [ 3.59%] __GI__mprotect
                └─ [ 0.68%] malloc_consolidate
                    └─ [ 0.32%] unlink_chunk.isra.2
                    └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI__libc_malloc
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                └─ [ 4.20%] sysmalloc
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                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
                        └─ [ 0.32%] unlink_chunk.isra.2
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    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI__libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI__mprotect
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
                        └─ [ 0.32%] unlink_chunk.isra.2

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```

└─ [ 3.69%] __memmove_evex_unaligned_erms
    └─ [ 0.24%] LOG_FIND_CURRENT_TDES
        └─ [ 1.90%] LOG_FIND_TDES
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI___libc_malloc
        └─ [ 6.52%] _int_malloc
            └─ [ 4.20%] sysmalloc
                └─ [ 3.59%] __GI___mprotect
            └─ [ 0.68%] malloc_consolidate
                └─ [ 0.32%] unlink_chunk.isra.2
            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 9.96%] log_zip
    └─ [ 8.84%] LZ4_compress_fast_extState
        └─ [ 1.72%] LZ4_read_ARCH
        └─ [ 0.78%] LZ4_read32
        └─ [ 0.47%] LZ4_initStream
            └─ [ 1.36%] __memset_evex_unaligned_erms
        └─ [ 0.21%] LZ4_writelE16
        └─ [ 1.06%] LZ4_resetStream_fast
            └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
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                └─ [ 0.68%] malloc_consolidate
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└─ [ 0.40%] prior_lsa_copy_redo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                └─ [ 0.68%] malloc_consolidate
                    └─ [ 0.32%] unlink_chunk.isra.2
                    └─ [ 0.32%] unlink_chunk.isra.2
            └─ [ 3.69%] __memmove_evex_unaligned_erms
        └─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 7.94%] prior_lsa_next_record_internal
    └─ [ 5.61%] __GI___pthread_mutex_lock
        └─ [ 2.75%] _lll_lock_wait
    └─ [ 4.27%] _pthread_mutex_unlock_usercnt
        └─ [ 2.37%] _lll_unlock_wake
    └─ [ 0.22%] prior_lsa_start_append
        └─ [ 0.22%] vacuum_get_log_blockid
    └─ [ 1.90%] LOG_FIND_TDES
└─ [ 0.34%] heap_mvcc_log_insert
    └─ [ 0.27%] logtb_get_current_mvccid
        └─ [ 1.90%] LOG_FIND_TDES
└─ [ 3.44%] spage_insert_at
    └─ [ 2.87%] spage_insert_data
        └─ [ 3.69%] __memmove_evex_unaligned_erms
            └─ [ 0.21%] pgbuf_set_dirty
    └─ [ 0.54%] spage_find_empty_slot_at
        └─ [ 0.35%] spage check space

```

```

└ [ 0.47%] spage_has_enough_total_space
  └ [ 0.60%] spage_get_total_saved_spaces
    └ [ 0.58%] spage_get_saved_spaces
      └ [ 0.45%] lf_hash_find
        └ [ 0.26%] lf_list_find
└ [ 1.99%] pgbuf_unfix
  └ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
    └ [ 4.27%] __pthread_mutex_unlock_usercnt
      └ [ 2.37%] __l1_unlock_wake
        └ [ 5.61%] __GI__pthread_mutex_lock
          └ [ 2.75%] __l1_lock_wait
            └ [ 0.27%] logtb_get_current_mvccid
              └ [ 1.90%] LOG_FIND_TDES
└ [ 10.84%] locator_add_or_remove_index_internal
  └ [ 7.17%] btree_insert
    └ [ 7.07%] btree_insert_internal
      └ [ 4.71%] btree_search_key_and_apply_functions
        └ [ 4.05%] btree_fix_root_for_insert
          └ [ 3.33%] logtb_tran_update_unique_stats
            └ [ 3.68%] log_append_undo_data2
              └ [ 3.66%] log_append_undo_crumbs
                └ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
                  └ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
                    └ [ 9.96%] log_zip
                      └ [ 8.84%] LZ4_compress_fast_extState
                        └ [ 1.72%] LZ4_read_ARCH
                          └ [ 0.78%] LZ4_read32
                            └ [ 0.47%] LZ4_initStream
                              └ [ 1.36%] __memset_evex_unaligned_erms
                                └ [ 0.21%] LZ4_writeLE16
                                  └ [ 1.06%] LZ4_resetStream_fast
                                    └ [ 1.36%] __memset_evex_unaligned_erms
└ [ 6.08%] cub_alloc
  └ [ 7.24%] __GI__libc_malloc
    └ [ 6.52%] _int_malloc
      └ [ 4.20%] sysmalloc
        └ [ 3.59%] __GI__mprotect
          └ [ 0.68%] malloc_consolidate
            └ [ 0.32%] unlink_chunk.isra.2
              └ [ 0.32%] unlink_chunk.isra.2
└ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
  └ [ 6.08%] cub_alloc
    └ [ 7.24%] __GI__libc_malloc
      └ [ 6.52%] _int_malloc
        └ [ 4.20%] sysmalloc
          └ [ 3.59%] __GI__mprotect
            └ [ 0.68%] malloc_consolidate
              └ [ 0.32%] unlink_chunk.isra.2
                └ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
  └ [ 6.08%] cub_alloc
    └ [ 7.24%] __GI__libc_malloc
      └ [ 6.52%] _int_malloc
        └ [ 4.20%] sysmalloc
          └ [ 3.59%] __GI__mprotect
            └ [ 0.68%] malloc_consolidate
              └ [ 0.32%] unlink_chunk.isra.2

```

```

└ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 0.24%] LOG_FIND_CURRENT_TDES
└ [ 1.90%] LOG_FIND_TDES
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 9.96%] log_zip
└ [ 8.84%] LZ4_compress_fast_extState
└ [ 1.72%] LZ4_read_ARCH
└ [ 0.78%] LZ4_read32
└ [ 0.47%] LZ4_initStream
└ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.21%] LZ4_writeLE16
└ [ 1.06%] LZ4_resetStream_fast
└ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.59%] prior_lsa_copy_undo_data_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.40%] prior_lsa_copy_redo_data_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 7.94%] prior_lsa_next_record_internal
└ [ 5.61%] __GI__pthread_mutex_lock
└ [ 2.75%] __lll_lock_wait
└ [ 4.27%] __pthread_mutex_unlock_usercnt
└ [ 2.37%] __lll_unlock_wake
└ [ 0.22%] prior_lsa_start_append
└ [ 0.22%] vacuum_get_log_blockid
└ [ 1.90%] LOG_FIND_TDES
└ [ 0.28%] logtb_tran_update_btid_unique_stats
└ [ 0.24%] logtb_tran_find_btid_stats
└ [ 0.34%] mht_get
└ [ 1.90%] LOG_FIND_TDES
└ [ 3.91%] pgbuf_fix_release
└ [ 1.56%] pgbuf_claim_bcb_for_fix
└ [ 1.33%] fileio_init_lsa_of_page
└ [ 1.35%] LSA_SET_NULL

```

```

└ [ 0.39%] pgbuf_search_hash_chain
  └ [ 0.40%] __GI___pthread_mutex_trylock
└ [ 0.30%] pgbuf_latch_bcb_upon_fix
  └ [ 4.27%] __pthread_mutex_unlock_usercnt
    └ [ 2.37%] __l1_unlock_wake
└ [ 1.99%] pgbuf_unfix
  └ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
    └ [ 4.27%] __pthread_mutex_unlock_usercnt
      └ [ 2.37%] __l1_unlock_wake
└ [ 5.61%] __GI___pthread_mutex_lock
  └ [ 2.75%] __l1_lock_wait
└ [ 2.19%] btree_split_node_and_advance
  └ [ 0.77%] btree_search_leaf_page
    └ [ 0.46%] btree_read_record_without_decompression
      └ [ 0.27%] btree_compare_key
        └ [ 0.46%] spage_get_record
          └ [ 0.32%] spage_find_slot
└ [ 0.68%] btree_search_nonleaf_page
  └ [ 0.46%] btree_read_record_without_decompression
    └ [ 0.77%] btree_search_leaf_page
      └ [ 0.27%] btree_compare_key
        └ [ 0.46%] spage_get_record
          └ [ 0.32%] spage_find_slot
└ [ 0.27%] btree_compare_key
  └ [ 0.46%] spage_get_record
    └ [ 0.32%] spage_find_slot
└ [ 3.91%] pgbuf_fix_release
  └ [ 1.56%] pgbuf_claim_bcb_for_fix
    └ [ 1.33%] fileio_init_lsa_of_page
      └ [ 1.35%] LSA_SET_NULL
└ [ 0.39%] pgbuf_search_hash_chain
  └ [ 0.40%] __GI___pthread_mutex_trylock
└ [ 0.30%] pgbuf_latch_bcb_upon_fix
  └ [ 4.27%] __pthread_mutex_unlock_usercnt
    └ [ 2.37%] __l1_unlock_wake
└ [ 5.46%] file_alloc
  └ [ 3.91%] pgbuf_fix_release
    └ [ 1.56%] pgbuf_claim_bcb_for_fix
      └ [ 1.33%] fileio_init_lsa_of_page
        └ [ 1.35%] LSA_SET_NULL
└ [ 0.39%] pgbuf_search_hash_chain
  └ [ 0.40%] __GI___pthread_mutex_trylock
└ [ 0.30%] pgbuf_latch_bcb_upon_fix
  └ [ 4.27%] __pthread_mutex_unlock_usercnt
    └ [ 2.37%] __l1_unlock_wake
└ [ 1.25%] file_perm_alloc
  └ [ 0.62%] log_append_undoredo_data2
    └ [ 19.52%] log_append_undoredo_crumbs
      └ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
        └ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
          └ [ 9.96%] log_zip
            └ [ 8.84%] LZ4_compress_fast_extState
              └ [ 1.72%] LZ4_read_ARCH
                └ [ 0.78%] LZ4_read32
                  └ [ 0.47%] LZ4_initStream
                    └ [ 1.36%] __memset_evex_unaligned_erms
                      └ [ 0.21%] LZ4_writeLE16

```

```

└ [ 1.06%] LZ4_resetStream_fast
  └ [ 1.36%] __memset_evex_unaligned_erms
└ [ 6.08%] cub_alloc
  └ [ 7.24%] __GI__libc_malloc
    └ [ 6.52%] _int_malloc
      └ [ 4.20%] sysmalloc
        └ [ 3.59%] __GI__mprotect
          └ [ 0.68%] malloc_consolidate
            └ [ 0.32%] unlink_chunk.isra.2
              └ [ 0.32%] unlink_chunk.isra.2
└ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
  └ [ 6.08%] cub_alloc
    └ [ 7.24%] __GI__libc_malloc
      └ [ 6.52%] _int_malloc
        └ [ 4.20%] sysmalloc
          └ [ 3.59%] __GI__mprotect
            └ [ 0.68%] malloc_consolidate
              └ [ 0.32%] unlink_chunk.isra.2
                └ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
  └ [ 6.08%] cub_alloc
    └ [ 7.24%] __GI__libc_malloc
      └ [ 6.52%] _int_malloc
        └ [ 4.20%] sysmalloc
          └ [ 3.59%] __GI__mprotect
            └ [ 0.68%] malloc_consolidate
              └ [ 0.32%] unlink_chunk.isra.2
                └ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 0.24%] LOG_FIND_CURRENT_TDES
  └ [ 1.90%] LOG_FIND_TDES
└ [ 6.08%] cub_alloc
  └ [ 7.24%] __GI__libc_malloc
    └ [ 6.52%] _int_malloc
      └ [ 4.20%] sysmalloc
        └ [ 3.59%] __GI__mprotect
          └ [ 0.68%] malloc_consolidate
            └ [ 0.32%] unlink_chunk.isra.2
              └ [ 0.32%] unlink_chunk.isra.2
└ [ 9.96%] log_zip
  └ [ 8.84%] LZ4_compress_fast_extState
    └ [ 1.72%] LZ4_read_ARCH
      └ [ 0.78%] LZ4_read32
        └ [ 0.47%] LZ4_initStream
          └ [ 1.36%] __memset_evex_unaligned_erms
            └ [ 0.21%] LZ4_writeLE16
              └ [ 1.06%] LZ4_resetStream_fast
                └ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.59%] prior_lsa_copy_undo_data_to_node
  └ [ 6.08%] cub_alloc
    └ [ 7.24%] __GI__libc_malloc
      └ [ 6.52%] _int_malloc
        └ [ 4.20%] sysmalloc
          └ [ 3.59%] __GI__mprotect
            └ [ 0.68%] malloc_consolidate
              └ [ 0.32%] unlink_chunk.isra.2
                └ [ 0.32%] unlink_chunk.isra.2

```

```

└ [ 0.40%] prior_lsa_copy_redo_data_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 7.94%] prior_lsa_next_record_internal
└ [ 5.61%] __GI__pthread_mutex_lock
└ [ 2.75%] _lll_lock_wait
└ [ 4.27%] _pthread_mutex_unlock_usercnt
└ [ 2.37%] _lll_unlock_wake
└ [ 0.22%] prior_lsa_start_append
└ [ 0.22%] vacuum_get_log_blockid
└ [ 1.90%] LOG_FIND_TDES
└ [ 7.88%] log_append_undoredo_data
└ [ 19.52%] log_append_undoredo_crumbs
└ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
└ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
└ [ 9.96%] log_zip
└ [ 8.84%] LZ4_compress_fast_extState
└ [ 1.72%] LZ4_read_ARCH
└ [ 0.78%] LZ4_read32
└ [ 0.47%] LZ4_initStream
└ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.21%] LZ4_writeLE16
└ [ 1.06%] LZ4_resetStream_fast
└ [ 1.36%] __memset_evex_unaligned_erms
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
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└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2

```



```

└ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 0.24%] LOG_FIND_CURRENT_TDES
└ [ 1.90%] LOG_FIND_TDES
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
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└ [ 8.84%] LZ4_compress_fast_extState
└ [ 1.72%] LZ4_read_ARCH
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└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
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└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.40%] prior_lsa_copy_redo_data_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
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└ [ 0.68%] malloc_consolidate
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└ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 7.94%] prior_lsa_next_record_internal
└ [ 5.61%] __GI__pthread_mutex_lock
└ [ 2.75%] __lll_lock_wait
└ [ 4.27%] __pthread_mutex_unlock_usercnt
└ [ 2.37%] __lll_unlock_wake
└ [ 0.22%] prior_lsa_start_append
└ [ 0.22%] vacuum_get_log_blockid
└ [ 1.90%] LOG_FIND_TDES
└ [ 0.88%] heap_vpid_init_new
└ [ 7.88%] log_append_undoredo_data
└ [ 19.52%] log_append_undoredo_crumbs
└ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
└ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
└ [ 9.96%] log_zip
└ [ 8.84%] LZ4_compress_fast_extState
└ [ 1.72%] LZ4_read_ARCH

```

```

└─ [ 0.78%] LZ4_read32
└─ [ 0.47%] LZ4_initStream
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.21%] LZ4_writeLE16
└─ [ 1.06%] LZ4_resetStream_fast
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
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└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
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└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
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└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 0.24%] LOG_FIND_CURRENT_TDES
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 6.08%] cub_alloc
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└─ [ 1.06%] LZ4_resetStream_fast
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc

```

```
└─ [ 0.39%] log_tdes::unlock_topop
    └─ [ 0.38%] cubpl::get_session
        └─ [ 0.24%] session_get_pl_session
            └─ [ 0.27%] thread get thread entry info
```

```

└─ [ 0.21%] cubthread::get_entry
└─ [ 7.94%] prior_lsa_next_record_internal
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __l1ll_lock_wait
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __l1ll_unlock_wake
└─ [ 0.22%] prior_lsa_start_append
└─ [ 0.22%] vacuum_get_log_blockid
└─ [ 1.99%] pgbuf_unfix
└─ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __l1ll_unlock_wake
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __l1ll_lock_wait
└─ [ 1.27%] heap_get_class_name_alloc_if_diff
└─ [ 0.53%] heap_get_class_record
└─ [ 0.37%] heap_get_last_version
└─ [ 0.35%] heap_sancache_end
└─ [ 0.36%] heap_sancache_quick_end
└─ [ 1.99%] pgbuf_unfix
└─ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __l1ll_unlock_wake
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __l1ll_lock_wait
└─ [ 0.20%] cub_strdup
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.93%] heap_attrinfo_end
└─ [ 1.00%] heap_classrepr_free
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __l1ll_unlock_wake
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __l1ll_lock_wait
└─ [ 1.10%] heap_attrinfo_start_with_index
└─ [ 0.96%] heap_classrepr_get
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __l1ll_lock_wait
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __l1ll_unlock_wake
└─ [ 0.40%] __GI___pthread_mutex_trylock
└─ [ 0.21%] heap_attrinfo_recache_attrepr
└─ [ 0.54%] heap_attrinfo_read_dbvalues
└─ [ 0.28%] heap_attrvalue_read
└─ [ 0.27%] logtb_get_current_mvccid
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 0.99%] locator_check_foreign_key
└─ [ 1.10%] heap_attrinfo_start_with_index
└─ [ 0.96%] heap_classrepr_get
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __l1ll_lock_wait

```

[illegible]

```
tbb::detail::d2::micro_queue< cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator< cubconn::connection::worker::message> >::prepare_page(unsigned long,
```

```
tbb::detail::d2::micro_queue< cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator< cubconn::connection::worker::message> >::prepare_page(unsigned long,
tbb::detail::d2::concurrent_queue_rep< cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator< cubconn::connection::worker::message> >&,
tbb::detail::d1::cache_aligned_allocator< tbb::detail::d2::micro_queue< cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator< cubconn::connection::worker::message> >::padded_page>,
tbb::detail::d2::micro_queue< cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator< cubconn::connection::worker::message> >::padded_page*&))::
```

```

{lambda()#1}::operator()
├ [ 0.82%] tbb::detail::r1::cache_aligned_allocate
│   └ [ 0.80%] _mid_memalign
│       └ [ 0.74%] _int_memalign
│           ├── [ 6.52%] _int_malloc
│           │   ├── [ 4.20%] sysmalloc
│           │   │   └ [ 3.59%] __GI___mprotect
│           │   └ [ 0.68%] malloc_consolidate
│           │       └ [ 0.32%] unlink_chunk.isra.2
│           └ [ 0.32%] unlink_chunk.isra.2
│               └ [ 0.47%] _int_free
│                   ├── [ 0.68%] malloc_consolidate
│                   │   └ [ 0.32%] unlink_chunk.isra.2
│                   └ [ 0.32%] unlink_chunk.isra.2
├ [ 0.69%] rmutex_lock
│   ├── [ 5.61%] __GI___pthread_mutex_lock
│   │   └ [ 2.75%] __lll_lock_wait
│   └ [ 0.27%] thread_get_thread_entry_info
│       └ [ 0.21%] cubthread::get_entry
├ [ 0.23%] rmutex_unlock
│   ├── [ 4.27%] __pthread_mutex_unlock_usercnt
│   │   └ [ 2.37%] __lll_unlock_wake
│   └ [ 0.27%] thread_get_thread_entry_info
│       └ [ 0.21%] cubthread::get_entry
├ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 0.29%] css_receive_data_from_client_with_timeout
    └ [ 0.45%] css_receive_data
        ├── [ 0.27%] css_return_queued_data_timeout
        │   ├── [ 0.69%] rmutex_lock
        │   │   ├── [ 5.61%] __GI___pthread_mutex_lock
        │   │   │   └ [ 2.75%] __lll_lock_wait
        │   │   └ [ 0.27%] thread_get_thread_entry_info
        │   │       └ [ 0.21%] cubthread::get_entry
        │   └ [ 0.23%] rmutex_unlock
        │       ├── [ 4.27%] __pthread_mutex_unlock_usercnt
        │       │   └ [ 2.37%] __lll_unlock_wake
        │       └ [ 0.27%] thread_get_thread_entry_info
        │           └ [ 0.21%] cubthread::get_entry
        └ [ 6.08%] cub_alloc
            ├── [ 7.24%] __GI___libc_malloc
            │   └ [ 6.52%] _int_malloc
            │       ├── [ 4.20%] sysmalloc
            │       │   └ [ 3.59%] __GI___mprotect
            │       └ [ 0.68%] malloc_consolidate
            │           └ [ 0.32%] unlink_chunk.isra.2
            └ [ 0.32%] unlink_chunk.isra.2
├ [ 1.89%] stran_server_commit
│   └ [ 1.87%] xtran_server_commit
│       └ [ 1.87%] log_commit
│           └ [ 1.85%] log_commit_local
│               └ [ 1.80%] lock_unlock_all
│                   ├── [ 0.99%] lock_remove_resource
│                   │   └ [ 0.98%] lf_hash_delete_already_locked
│                   │       └ [ 0.94%] lf_list_delete
│                   │           ├── [ 2.48%] lock_res_key_compare
│                   │           └ [ 0.56%] lf_freelist_retire
│                   └ [ 0.40%] lf_freelist_transport

```



```

└─ [ 0.33%] lock_dealloc_entry
    └─ [ 0.47%] _int_free
        └─ [ 0.68%] malloc_consolidate
            └─ [ 0.32%] unlink_chunk.isra.2
unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.81%] lock_internal_perform_unlock_object
    └─ [ 0.56%] lf_freelist_retire
        └─ [ 0.40%] lf_freelist_transport
            └─ [ 0.33%] lock_dealloc_entry
                └─ [ 0.47%] _int_free
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
                            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 5.61%] __GI__pthread_mutex_lock
    └─ [ 2.75%] __lll_lock_wait
└─ [ 0.44%] slogwr_get_log_pages
    └─ [ 0.44%] xlogwr_get_log_pages
└─ [ 1.18%] css_request_release_packet
    └─ [ 1.15%] cubconn::connection::worker::enqueue
        └─ [ 0.82%]
tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::prepare_page(unsigned long,
tbb::detail::d2::concurrent_queue_rep<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >&,
tbb::detail::d1::cache_aligned_allocator<tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::padded_page>,
tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::padded_page*&):
{lambda()#1}::operator()
└─ [ 0.82%] tbb::detail::r1::cache_aligned_allocate
    └─ [ 0.80%] _mid_memalign
        └─ [ 0.74%] _int_memalign
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI__mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
                                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.47%] _int_free
    └─ [ 0.68%] malloc_consolidate
        └─ [ 0.32%] unlink_chunk.isra.2
            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.69%] rmutex_lock
    └─ [ 5.61%] __GI__pthread_mutex_lock
        └─ [ 2.75%] __lll_lock_wait
└─ [ 0.27%] thread_get_thread_entry_info
    └─ [ 0.21%] cubthread::get_entry
└─ [ 0.23%] rmutex_unlock
    └─ [ 4.27%] __pthread_mutex_unlock_usercnt
        └─ [ 2.37%] __lll_unlock_wake
└─ [ 0.27%] thread_get_thread_entry_info
    └─ [ 0.21%] cubthread::get_entry
└─ [ 0.91%] css_internal_request_handler
    └─ [ 0.63%] css_return_queued_request
        └─ [ 1.18%] css_request_release_packet
            └─ [ 1.15%] cubconn::connection::worker::enqueue

```

```
tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::prepare_page(unsigned long,
tbb::detail::d2::concurrent_queue_rep<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >&,
tbb::detail::d1::cache_aligned_allocator<tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::padded_page>,
tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::padded_page*&))::
{lambda()#1}::operator()
```

```
[ 0.82%] tbb::detail::r1::cache_aligned_allocate
└─ [ 0.80%] _mid_memalign
   └─ [ 0.74%] _int_memalign
      ├── [ 6.52%] _int_malloc
      │   ├── [ 4.20%] sysmalloc
      │   │   └─ [ 3.59%] __GI___mprotect
      │   ├── [ 0.68%] malloc_consolidate
      │   │   └─ [ 0.32%] unlink_chunk.isra.2
      │   └─ [ 0.32%] unlink_chunk.isra.2
      └─ [ 0.47%] _int_free
         ├── [ 0.68%] malloc_consolidate
         │   └─ [ 0.32%] unlink_chunk.isra.2
         └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.69%] rmutex_lock
   ├── [ 5.61%] __GI___pthread_mutex_lock
   │   └─ [ 2.75%] __lll_lock_wait
   └─ [ 0.27%] thread_get_thread_entry_info
       └─ [ 0.21%] cubthread::get_entry
└─ [ 0.23%] rmutex_unlock
   ├── [ 4.27%] __pthread_mutex_unlock_usercnt
   │   └─ [ 2.37%] __lll_unlock_wake
   └─ [ 0.27%] thread_get_thread_entry_info
       └─ [ 0.21%] cubthread::get_entry
└─ [ 0.69%] rmutex_lock
   ├── [ 5.61%] __GI___pthread_mutex_lock
   │   └─ [ 2.75%] __lll_lock_wait
   └─ [ 0.27%] thread_get_thread_entry_info
       └─ [ 0.21%] cubthread::get_entry
└─ [ 0.45%] css_receive_data
   ├── [ 0.27%] css_return_queued_data_timeout
   │   └─ [ 0.69%] rmutex_lock
   │       ├── [ 5.61%] __GI___pthread_mutex_lock
   │       │   └─ [ 2.75%] __lll_lock_wait
   │       └─ [ 0.27%] thread_get_thread_entry_info
   │           └─ [ 0.21%] cubthread::get_entry
   └─ [ 0.23%] rmutex_unlock
       ├── [ 4.27%] __pthread_mutex_unlock_usercnt
       │   └─ [ 2.37%] __lll_unlock_wake
       └─ [ 0.27%] thread_get_thread_entry_info
           └─ [ 0.21%] cubthread::get_entry
└─ [ 6.08%] cub_alloc
   └─ [ 7.24%] __GI___libc_malloc
       └─ [ 6.52%] _int_malloc
           ├── [ 4.20%] sysmalloc
           │   └─ [ 3.59%] __GI___mprotect
           ├── [ 0.68%] malloc_consolidate
           │   └─ [ 0.32%] unlink chunk.isra.2
```

```

└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 2.74%] cubthread::worker_pool::core::worker::get_new_task
└─ [ 2.52%] __pthread_cond_timedwait
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __l1_unlock_wake
└─ [ 10.84%] locator_add_or_remove_index_internal
└─ [ 7.17%] btree_insert
└─ [ 7.07%] btree_insert_internal
└─ [ 4.71%] btree_search_key_and_apply_functions
└─ [ 4.05%] btree_fix_root_for_insert
└─ [ 3.33%] logtb_tran_update_unique_stats
└─ [ 3.68%] log_append_undo_data2
└─ [ 3.66%] log_append_undo_crumbs
└─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
└─ [ 11.92%] prior_lsa_gen_undoedo_record_from_crumbs
└─ [ 9.96%] log_zip
└─ [ 8.84%] LZ4_compress_fast_extState
└─ [ 1.72%] LZ4_read_ARCH
└─ [ 0.78%] LZ4_read32
└─ [ 0.47%] LZ4_initStream
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.21%] LZ4_writeLE16
└─ [ 1.06%] LZ4_resetStream_fast
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI__libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI__mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI__libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI__mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI__libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI__mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 0.24%] LOG_FIND_CURRENT_TDES
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI__libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc

```

```

└─ [ 3.59%] __GI___mprotect
    └─ [ 0.68%] malloc_consolidate
        └─ [ 0.32%] unlink_chunk.isra.2
            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 9.96%] log_zip
    └─ [ 8.84%] LZ4_compress_fast_extState
        └─ [ 1.72%] LZ4_read_ARCH
        └─ [ 0.78%] LZ4_read32
        └─ [ 0.47%] LZ4_initStream
            └─ [ 1.36%] __memset_evex_unaligned_erms
            └─ [ 0.21%] LZ4_writeLE16
        └─ [ 1.06%] LZ4_resetStream_fast
            └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
                        └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.40%] prior_lsa_copy_redo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
                        └─ [ 0.32%] unlink_chunk.isra.2
        └─ [ 3.69%] __memmove_evex_unaligned_erms
    └─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 7.94%] prior_lsa_next_record_internal
    └─ [ 5.61%] __GI___pthread_mutex_lock
        └─ [ 2.75%] __lll_lock_wait
    └─ [ 4.27%] __pthread_mutex_unlock_usercnt
        └─ [ 2.37%] __lll_unlock_wake
    └─ [ 0.22%] prior_lsa_start_append
        └─ [ 0.22%] vacuum_get_log_blockid
    └─ [ 1.90%] LOG_FIND_TDES
└─ [ 0.28%] logtb_tran_update_btid_unique_stats
    └─ [ 0.24%] logtb_tran_find_btid_stats
        └─ [ 0.34%] mht_get
        └─ [ 1.90%] LOG_FIND_TDES
└─ [ 3.91%] pgbuf_fix_release
    └─ [ 1.56%] pgbuf_claim_bcb_for_fix
        └─ [ 1.33%] fileio_init_lsa_of_page
            └─ [ 1.35%] LSA_SET_NULL
    └─ [ 0.39%] pgbuf_search_hash_chain
        └─ [ 0.40%] __GI___pthread_mutex_trylock
    └─ [ 0.30%] pgbuf_latch_bcb_upon_fix
        └─ [ 4.27%] __pthread_mutex_unlock_usercnt
            └─ [ 2.37%] __lll_unlock_wake
└─ [ 1.99%] pgbuf_unfix
    └─ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
        └─ [ 4.27%] __pthread_mutex_unlock_usercnt

```

```

└─ [ 2.37%] __lll_unlock_wake
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __lll_lock_wait
└─ [ 2.19%] btree_split_node_and_advance
└─ [ 0.77%] btree_search_leaf_page
└─ [ 0.46%] btree_read_record_without_decompression
└─ [ 0.27%] btree_compare_key
└─ [ 0.46%] spage_get_record
└─ [ 0.32%] spage_find_slot
└─ [ 0.68%] btree_search_nonleaf_page
└─ [ 0.46%] btree_read_record_without_decompression
└─ [ 0.77%] btree_search_leaf_page
└─ [ 0.27%] btree_compare_key
└─ [ 0.46%] spage_get_record
└─ [ 0.32%] spage_find_slot
└─ [ 0.27%] btree_compare_key
└─ [ 0.46%] spage_get_record
└─ [ 0.32%] spage_find_slot
└─ [ 3.91%] pgbuf_fix_release
└─ [ 1.56%] pgbuf_claim_bcb_for_fix
└─ [ 1.33%] fileio_init_lsa_of_page
└─ [ 1.35%] LSA_SET_NULL
└─ [ 0.39%] pgbuf_search_hash_chain
└─ [ 0.40%] __GI___pthread_mutex_trylock
└─ [ 0.30%] pgbuf_latch_bcb_upon_fix
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __lll_unlock_wake
└─ [ 5.46%] file_alloc
└─ [ 3.91%] pgbuf_fix_release
└─ [ 1.56%] pgbuf_claim_bcb_for_fix
└─ [ 1.33%] fileio_init_lsa_of_page
└─ [ 1.35%] LSA_SET_NULL
└─ [ 0.39%] pgbuf_search_hash_chain
└─ [ 0.40%] __GI___pthread_mutex_trylock
└─ [ 0.30%] pgbuf_latch_bcb_upon_fix
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __lll_unlock_wake
└─ [ 1.25%] file_perm_alloc
└─ [ 0.62%] log_append_undoredo_data2
└─ [ 19.52%] log_append_undoredo_crumbs
└─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
└─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
└─ [ 9.96%] log_zip
└─ [ 8.84%] LZ4_compress_fast_extState
└─ [ 1.72%] LZ4_read_ARCH
└─ [ 0.78%] LZ4_read32
└─ [ 0.47%] LZ4_initStream
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.21%] LZ4_writeLE16
└─ [ 1.06%] LZ4_resetStream_fast
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc consolidate

```

```

└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 0.24%] LOG_FIND_CURRENT_TDES
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 9.96%] log_zip
└─ [ 8.84%] LZ4_compress_fast_extState
└─ [ 1.72%] LZ4_read_ARCH
└─ [ 0.78%] LZ4_read32
└─ [ 0.47%] LZ4_initStream
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.21%] LZ4_writeLE16
└─ [ 1.06%] LZ4_resetStream_fast
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.40%] prior_lsa_copy_redo_data_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2

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└ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 7.94%] prior_lsa_next_record_internal
└ [ 5.61%] __GI__pthread_mutex_lock
└ [ 2.75%] __lll_lock_wait
└ [ 4.27%] __pthread_mutex_unlock_usercnt
└ [ 2.37%] __lll_unlock_wake
└ [ 0.22%] prior_lsa_start_append
└ [ 0.22%] vacuum_get_log_blockid
└ [ 1.90%] LOG_FIND_TDES
└ [ 7.88%] log_append_undoredo_data
└ [ 19.52%] log_append_undoredo_crumbs
└ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
└ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
└ [ 9.96%] log_zip
└ [ 8.84%] LZ4_compress_fast_extState
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└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
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└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 0.24%] LOG_FIND_CURRENT_TDES
└ [ 1.90%] LOG_FIND_TDES
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc

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└─ [ 3.59%] __GI___mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 9.96%] log_zip
└─ [ 8.84%] LZ4_compress_fast_extState
└─ [ 1.72%] LZ4_read_ARCH
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└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.40%] prior_lsa_copy_redo_data_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
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└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 7.94%] prior_lsa_next_record_internal
└─ [ 5.61%] __GI___pthread_mutex_lock
└─ [ 2.75%] __lll_lock_wait
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
└─ [ 2.37%] __lll_unlock_wake
└─ [ 0.22%] prior_lsa_start_append
└─ [ 0.22%] vacuum_get_log_blockid
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 0.88%] heap_vpid_init_new
└─ [ 7.88%] log_append_undoredo_data
└─ [ 19.52%] log_append_undoredo_crumbs
└─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
└─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
└─ [ 9.96%] log_zip
└─ [ 8.84%] LZ4_compress_fast_extState
└─ [ 1.72%] LZ4_read_ARCH
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└─ [ 0.47%] LZ4_initStream
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└─ [ 0.21%] LZ4_writeLE16
└─ [ 1.06%] LZ4_resetStream_fast
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI___libc_malloc

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└ [ 6.52%] _int_malloc
  └ [ 4.20%] sysmalloc
    └ [ 3.59%] __GI___mprotect
      └ [ 0.68%] malloc_consolidate
        └ [ 0.32%] unlink_chunk.isra.2
          └ [ 0.32%] unlink_chunk.isra.2
            └ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
              └ [ 6.08%] cub_alloc
                └ [ 7.24%] __GI___libc_malloc
                  └ [ 6.52%] _int_malloc
                    └ [ 4.20%] sysmalloc
                      └ [ 3.59%] __GI___mprotect
                        └ [ 0.68%] malloc_consolidate
                          └ [ 0.32%] unlink_chunk.isra.2
                            └ [ 0.32%] unlink_chunk.isra.2
                              └ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
                                └ [ 6.08%] cub_alloc
                                  └ [ 7.24%] __GI___libc_malloc
                                    └ [ 6.52%] _int_malloc
                                      └ [ 4.20%] sysmalloc
                                        └ [ 3.59%] __GI___mprotect
                                          └ [ 0.68%] malloc_consolidate
                                            └ [ 0.32%] unlink_chunk.isra.2
                                              └ [ 0.32%] unlink_chunk.isra.2
                                                └ [ 3.69%] __memmove_evex_unaligned_erms
                                                  └ [ 0.24%] LOG_FIND_CURRENT_TDES
                                                    └ [ 1.90%] LOG_FIND_TDES
                                                      └ [ 6.08%] cub_alloc
                                                        └ [ 7.24%] __GI___libc_malloc
                                                          └ [ 6.52%] _int_malloc
                                                            └ [ 4.20%] sysmalloc
                                                              └ [ 3.59%] __GI___mprotect
                                                                └ [ 0.68%] malloc_consolidate
                                                                  └ [ 0.32%] unlink_chunk.isra.2
                                                                    └ [ 0.32%] unlink_chunk.isra.2
                                                                      └ [ 9.96%] log_zip
                                                                        └ [ 8.84%] LZ4_compress_fast_extState
                                                                          └ [ 1.72%] LZ4_read_ARCH
                                                                            └ [ 0.78%] LZ4_read32
                                                                              └ [ 0.47%] LZ4_initStream
                                                                                └ [ 1.36%] __memset_evex_unaligned_erms
                                                                                  └ [ 0.21%] LZ4_writeLE16
                                                                                    └ [ 1.06%] LZ4_resetStream_fast
                                                                                      └ [ 1.36%] __memset_evex_unaligned_erms
                                                                                        └ [ 0.59%] prior_lsa_copy_undo_data_to_node
                                                                                          └ [ 6.08%] cub_alloc
                                                                                            └ [ 7.24%] __GI___libc_malloc
                                                                                              └ [ 6.52%] _int_malloc
                                                                                                └ [ 4.20%] sysmalloc
                                                                                                  └ [ 3.59%] __GI___mprotect
                                                                                                    └ [ 0.68%] malloc_consolidate
                                                                                                      └ [ 0.32%] unlink_chunk.isra.2
                                                                                                        └ [ 0.32%] unlink_chunk.isra.2
                                                                                                          └ [ 0.40%] prior_lsa_copy_redo_data_to_node
                                                                                                            └ [ 6.08%] cub_alloc
                                                                                                              └ [ 7.24%] __GI___libc_malloc
                                                                                                                └ [ 6.52%] _int_malloc

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└─ [ 4.20%] sysmalloc
    └─ [ 3.59%] __GI__mprotect
    └─ [ 0.68%] malloc_consolidate
    └─ [ 0.32%] unlink_chunk.isra.2
    └─ [ 0.32%] unlink_chunk.isra.2
    └─ [ 3.69%] __memmove_evex_unaligned_erms
    └─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 7.94%] prior_lsa_next_record_internal
    └─ [ 5.61%] __GI__pthread_mutex_lock
    └─ [ 2.75%] __lll_lock_wait
    └─ [ 4.27%] __pthread_mutex_unlock_usercnt
    └─ [ 2.37%] __lll_unlock_wake
    └─ [ 0.22%] prior_lsa_start_append
    └─ [ 0.22%] vacuum_get_log_blockid
    └─ [ 1.90%] LOG_FIND_TDES
└─ [ 0.23%] spage_insert
└─ [ 0.21%] pgbuf_set_dirty
└─ [ 0.64%] log_sysop_end_logical_undo
    └─ [ 1.17%] log_sysop_commit_internal
        └─ [ 0.99%] log_append_sysop_end
            └─ [ 7.94%] prior_lsa_next_record_internal
                └─ [ 5.61%] __GI__pthread_mutex_lock
                └─ [ 2.75%] __lll_lock_wait
                └─ [ 4.27%] __pthread_mutex_unlock_usercnt
                └─ [ 2.37%] __lll_unlock_wake
                └─ [ 0.22%] prior_lsa_start_append
                └─ [ 0.22%] vacuum_get_log_blockid
            └─ [ 0.45%] prior_lsa_alloc_and_copy_data
                └─ [ 6.08%] cub_alloc
                └─ [ 7.24%] __GI__libc_malloc
                └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                └─ [ 3.59%] __GI__mprotect
                └─ [ 0.68%] malloc_consolidate
                └─ [ 0.32%] unlink_chunk.isra.2
                └─ [ 0.32%] unlink_chunk.isra.2
            └─ [ 0.59%] prior_lsa_copy_undo_data_to_node
                └─ [ 6.08%] cub_alloc
                └─ [ 7.24%] __GI__libc_malloc
                └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                └─ [ 3.59%] __GI__mprotect
                └─ [ 0.68%] malloc_consolidate
                └─ [ 0.32%] unlink_chunk.isra.2
                └─ [ 0.32%] unlink_chunk.isra.2
        └─ [ 0.39%] log_tdes::unlock_topop
            └─ [ 0.38%] cubpl::get_session
                └─ [ 0.24%] session_get_pl_session
                └─ [ 0.27%] thread_get_thread_entry_info
                └─ [ 0.21%] cubthread::get_entry
└─ [ 7.94%] prior_lsa_next_record_internal
    └─ [ 5.61%] __GI__pthread_mutex_lock
    └─ [ 2.75%] __lll_lock_wait
    └─ [ 4.27%] __pthread_mutex_unlock_usercnt
    └─ [ 2.37%] __lll_unlock_wake
    └─ [ 0.22%] prior_lsa_start_append
    └─ [ 0.22%] vacuum_get_log_blockid

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└ [ 1.99%] pgbuf_unfix
  └ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
    └ [ 4.27%] __pthread_mutex_unlock_usercnt
      └ [ 2.37%] __l1_unlock_wake
        └ [ 5.61%] __GI__pthread_mutex_lock
          └ [ 2.75%] __l1_lock_wait
            └ [ 1.27%] heap_get_class_name_alloc_if_diff
              └ [ 0.53%] heap_get_class_record
                └ [ 0.37%] heap_get_last_version
                  └ [ 0.35%] heap_sancache_end
                    └ [ 0.36%] heap_sancache_quick_end
                      └ [ 1.99%] pgbuf_unfix
                        └ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
                          └ [ 4.27%] __pthread_mutex_unlock_usercnt
                            └ [ 2.37%] __l1_unlock_wake
                              └ [ 5.61%] __GI__pthread_mutex_lock
                                └ [ 2.75%] __l1_lock_wait
                                  └ [ 0.20%] cub_strdup
                                    └ [ 6.08%] cub_alloc
                                      └ [ 7.24%] __GI__libc_malloc
                                        └ [ 6.52%] _int_malloc
                                          └ [ 4.20%] sysmalloc
                                            └ [ 3.59%] __GI__mprotect
                                              └ [ 0.68%] malloc_consolidate
                                                └ [ 0.32%] unlink_chunk.isra.2
                                                  └ [ 0.32%] unlink_chunk.isra.2
                                                    └ [ 0.93%] heap_attrinfo_end
                                                      └ [ 1.00%] heap_classrepr_free
                                                        └ [ 4.27%] __pthread_mutex_unlock_usercnt
                                                          └ [ 2.37%] __l1_unlock_wake
                                                            └ [ 5.61%] __GI__pthread_mutex_lock
                                                              └ [ 2.75%] __l1_lock_wait
                                                                └ [ 1.10%] heap_attrinfo_start_with_index
                                                                  └ [ 0.96%] heap_classrepr_get
                                                                    └ [ 5.61%] __GI__pthread_mutex_lock
                                                                      └ [ 2.75%] __l1_lock_wait
                                                                        └ [ 4.27%] __pthread_mutex_unlock_usercnt
                                                                          └ [ 2.37%] __l1_unlock_wake
                                                                            └ [ 0.40%] __GI__pthread_mutex_trylock
                                                                              └ [ 0.21%] heap_attrinfo_recache_attrepr
                                                                                └ [ 0.54%] heap_attrinfo_read_dbvalues
                                                                                  └ [ 0.28%] heap_attrvalue_read
                                                                                    └ [ 0.27%] logtb_get_current_mvccid
                                                                                      └ [ 1.90%] LOG_FIND_TDES
                                                                                        └ [ 11.44%] heap_log_insert_physical
                                                                                          └ [ 19.52%] log_append_undoredo_crumbs
                                                                                            └ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
                                                                                              └ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
                                                                                                └ [ 9.96%] log_zip
                                                                                                  └ [ 8.84%] LZ4_compress_fast_extState
                                                                                                    └ [ 1.72%] LZ4_read_ARCH
                                                                                                      └ [ 0.78%] LZ4_read32
                                                                                                        └ [ 0.47%] LZ4_initStream
                                                                                                          └ [ 1.36%] __memset_evex_unaligned_erms
                                                                                                            └ [ 0.21%] LZ4_writeLE16
                                                                                                              └ [ 1.06%] LZ4_resetStream_fast
                                                                                                                └ [ 1.36%] __memset_evex_unaligned_erms

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└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI___libc_malloc
        └─ [ 6.52%] _int_malloc
            ├── [ 4.20%] sysmalloc
            │   └─ [ 3.59%] __GI___mprotect
            ├── [ 0.68%] malloc_consolidate
            │   └─ [ 0.32%] unlink_chunk.isra.2
            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                ├── [ 4.20%] sysmalloc
                │   └─ [ 3.59%] __GI___mprotect
                ├── [ 0.68%] malloc_consolidate
                │   └─ [ 0.32%] unlink_chunk.isra.2
                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                ├── [ 4.20%] sysmalloc
                │   └─ [ 3.59%] __GI___mprotect
                ├── [ 0.68%] malloc_consolidate
                │   └─ [ 0.32%] unlink_chunk.isra.2
                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 0.24%] LOG_FIND_CURRENT_TDES
    └─ [ 1.90%] LOG_FIND_TDES
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI___libc_malloc
        └─ [ 6.52%] _int_malloc
            ├── [ 4.20%] sysmalloc
            │   └─ [ 3.59%] __GI___mprotect
            ├── [ 0.68%] malloc_consolidate
            │   └─ [ 0.32%] unlink_chunk.isra.2
            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 9.96%] log_zip
    ├── [ 8.84%] LZ4_compress_fast_extState
    │   ├── [ 1.72%] LZ4_read_ARCH
    │   ├── [ 0.78%] LZ4_read32
    │   ├── [ 0.47%] LZ4_initStream
    │   │   └─ [ 1.36%] __memset_evex_unaligned_erms
    │   └─ [ 0.21%] LZ4_writeLE16
    ├── [ 1.06%] LZ4_resetStream_fast
    │   └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                ├── [ 4.20%] sysmalloc
                │   └─ [ 3.59%] __GI___mprotect
                ├── [ 0.68%] malloc_consolidate
                │   └─ [ 0.32%] unlink_chunk.isra.2
                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.40%] prior_lsa_copy_redo_data_to_node
    └─ [ 6.08%] cub_alloc

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└─ [ 7.24%] __GI___libc_malloc
    └─ [ 6.52%] _int_malloc
        └─ [ 4.20%] sysmalloc
            └─ [ 3.59%] __GI___mprotect
                └─ [ 0.68%] malloc_consolidate
                    └─ [ 0.32%] unlink_chunk.isra.2
                        └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 7.94%] prior_lsa_next_record_internal
    └─ [ 5.61%] __GI___pthread_mutex_lock
        └─ [ 2.75%] _lll_lock_wait
            └─ [ 4.27%] __pthread_mutex_unlock_usercnt
                └─ [ 2.37%] _lll_unlock_wake
                    └─ [ 0.22%] prior_lsa_start_append
                        └─ [ 0.22%] vacuum_get_log_blockid
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 0.34%] heap_mvcc_log_insert
    └─ [ 0.27%] logtb_get_current_mvccid
        └─ [ 1.90%] LOG_FIND_TDES
└─ [ 3.75%] btree_key_insert_new_key
    └─ [ 7.88%] log_append_undoredo_data
        └─ [ 19.52%] log_append_undoredo_crumbs
            └─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
                └─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
                    └─ [ 9.96%] log_zip
                        └─ [ 8.84%] LZ4_compress_fast_extState
                            └─ [ 1.72%] LZ4_read_ARCH
                                └─ [ 0.78%] LZ4_read32
                                    └─ [ 0.47%] LZ4_initStream
                                        └─ [ 1.36%] __memset_evex_unaligned_erms
                                            └─ [ 0.21%] LZ4_writeLE16
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    └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI___libc_malloc
        └─ [ 6.52%] _int_malloc
            └─ [ 4.20%] sysmalloc
                └─ [ 3.59%] __GI___mprotect
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
                            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
                                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect

```

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└─ [ 0.68%] malloc_consolidate
    └─ [ 0.32%] unlink_chunk.isra.2
        └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
    └─ [ 0.24%] LOG_FIND_CURRENT_TDES
        └─ [ 1.90%] LOG_FIND_TDES
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI__libc_malloc
        └─ [ 6.52%] _int_malloc
            └─ [ 4.20%] sysmalloc
                └─ [ 3.59%] __GI__mprotect
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
                            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 9.96%] log_zip
    └─ [ 8.84%] LZ4_compress_fast_extState
        └─ [ 1.72%] LZ4_read_ARCH
            └─ [ 0.78%] LZ4_read32
                └─ [ 0.47%] LZ4_initStream
                    └─ [ 1.36%] __memset_evex_unaligned_erms
                        └─ [ 0.21%] LZ4_writeLE16
└─ [ 1.06%] LZ4_resetStream_fast
    └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI__libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI__mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
                                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.40%] prior_lsa_copy_redo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI__libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI__mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
                                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
    └─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 7.94%] prior_lsa_next_record_internal
    └─ [ 5.61%] __GI__pthread_mutex_lock
        └─ [ 2.75%] __lll_lock_wait
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
    └─ [ 2.37%] __lll_unlock_wake
└─ [ 0.22%] prior_lsa_start_append
    └─ [ 0.22%] vacuum_get_log_blockid
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 3.44%] spage_insert_at
    └─ [ 2.87%] spage_insert_data
        └─ [ 3.69%] __memmove_evex_unaligned_erms
            └─ [ 0.21%] pgbuf_set_dirty
└─ [ 0.54%] spage_find_empty_slot_at
    └─ [ 0.35%] spage_check_space

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└─ [ 0.47%] spage_has_enough_total_space
    └─ [ 0.60%] spage_get_total_saved_spaces
        └─ [ 0.58%] spage_get_saved_spaces
            └─ [ 0.45%] lf_hash_find
                └─ [ 0.26%] lf_list_find
└─ [ 11.94%] heap_stats_find_best_page
    └─ [ 6.89%] heap_vpid_alloc
        └─ [ 5.46%] file_alloc
            └─ [ 3.91%] pgbuf_fix_release
                └─ [ 1.56%] pgbuf_claim_bcb_for_fix
                    └─ [ 1.33%] fileio_init_lsa_of_page
                        └─ [ 1.35%] LSA_SET_NULL
                            └─ [ 0.39%] pgbuf_search_hash_chain
                                └─ [ 0.40%] __GI___pthread_mutex_trylock
                                    └─ [ 0.30%] pgbuf_latch_bcb_upon_fix
                                        └─ [ 4.27%] __pthread_mutex_unlock_usercnt
                                            └─ [ 2.37%] __lll_unlock_wake
└─ [ 1.25%] file_perm_alloc
    └─ [ 0.62%] log_append_undoredo_data2
        └─ [ 19.52%] log_append_undoredo_crumbs
            └─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
                └─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
                    └─ [ 9.96%] log_zip
                        └─ [ 8.84%] LZ4_compress_fast_extState
                            └─ [ 1.72%] LZ4_read_ARCH
                                └─ [ 0.78%] LZ4_read32
                                    └─ [ 0.47%] LZ4_initStream
                                        └─ [ 1.36%] __memset_evex_unaligned_erms
                                            └─ [ 0.21%] LZ4_writeLE16
                                                └─ [ 1.06%] LZ4_resetStream_fast
                                                    └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI___libc_malloc
        └─ [ 6.52%] _int_malloc
            └─ [ 4.20%] sysmalloc
                └─ [ 3.59%] __GI___mprotect
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2

```

```

└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 0.24%] LOG_FIND_CURRENT_TDES
└ [ 1.90%] LOG_FIND_TDES
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 9.96%] log_zip
└ [ 8.84%] LZ4_compress_fast_extState
└ [ 1.72%] LZ4_read_ARCH
└ [ 0.78%] LZ4_read32
└ [ 0.47%] LZ4_initStream
└ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.21%] LZ4_writeLE16
└ [ 1.06%] LZ4_resetStream_fast
└ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.59%] prior_lsa_copy_undo_data_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.40%] prior_lsa_copy_redo_data_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI__libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI__mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 7.94%] prior_lsa_next_record_internal
└ [ 5.61%] __GI__pthread_mutex_lock
└ [ 2.75%] _lll_lock_wait
└ [ 4.27%] __pthread_mutex_unlock_usercnt
└ [ 2.37%] _lll_unlock_wake
└ [ 0.22%] prior_lsa_start_append
└ [ 0.22%] vacuum_get_log_blockid
└ [ 1.90%] LOG_FIND_TDES
└ [ 7.88%] log_append_undoredo_data
└ [ 19.52%] log_append_undoredo_crumbs
└ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
└ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
└ [ 9.96%] log_zip
└ [ 8.84%] LZ4_compress_fast_extState
└ [ 1.72%] LZ4_read_ARCH
└ [ 0.78%] LZ4_read32
└ [ 0.47%] LZ4_initStream

```



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└─ [ 1.36%] __memset_evex_unaligned_erms
    └─ [ 0.21%] LZ4_writeLE16
        └─ [ 1.06%] LZ4_resetStream_fast
            └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI__libc_malloc
        └─ [ 6.52%] _int_malloc
            ├── [ 4.20%] sysmalloc
            │   ├── [ 3.59%] __GI__mprotect
            │   ├── [ 0.68%] malloc_consolidate
            │   │   ├── [ 0.32%] unlink_chunk.isra.2
            │   │   └─ [ 0.32%] unlink_chunk.isra.2
            └─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
                └─ [ 6.08%] cub_alloc
                    └─ [ 7.24%] __GI__libc_malloc
                        └─ [ 6.52%] _int_malloc
                            ├── [ 4.20%] sysmalloc
                            │   ├── [ 3.59%] __GI__mprotect
                            │   ├── [ 0.68%] malloc_consolidate
                            │   │   ├── [ 0.32%] unlink_chunk.isra.2
                            │   │   └─ [ 0.32%] unlink_chunk.isra.2
                            └─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
                                └─ [ 6.08%] cub_alloc
                                    └─ [ 7.24%] __GI__libc_malloc
                                        └─ [ 6.52%] _int_malloc
                                            ├── [ 4.20%] sysmalloc
                                            │   ├── [ 3.59%] __GI__mprotect
                                            │   ├── [ 0.68%] malloc_consolidate
                                            │   │   ├── [ 0.32%] unlink_chunk.isra.2
                                            │   │   └─ [ 0.32%] unlink_chunk.isra.2
                                            └─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 0.24%] LOG_FIND_CURRENT_TDES
    └─ [ 1.90%] LOG_FIND_TDES
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI__libc_malloc
        └─ [ 6.52%] _int_malloc
            ├── [ 4.20%] sysmalloc
            │   ├── [ 3.59%] __GI__mprotect
            │   ├── [ 0.68%] malloc_consolidate
            │   │   ├── [ 0.32%] unlink_chunk.isra.2
            │   │   └─ [ 0.32%] unlink_chunk.isra.2
            └─ [ 9.96%] log_zip
                ├── [ 8.84%] LZ4_compress_fast_extState
                │   ├── [ 1.72%] LZ4_read_ARCH
                │   ├── [ 0.78%] LZ4_read32
                │   ├── [ 0.47%] LZ4_initStream
                │   │   └─ [ 1.36%] __memset_evex_unaligned_erms
                │   └─ [ 0.21%] LZ4_writeLE16
                └─ [ 1.06%] LZ4_resetStream_fast
                    └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI__libc_malloc
            └─ [ 6.52%] _int_malloc
                ├── [ 4.20%] sysmalloc
                │   ├── [ 3.59%] __GI__mprotect
                │   ├── [ 0.68%] malloc_consolidate
```

```

└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.40%] prior_lsa_copy_redo_data_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI__libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI__mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 7.94%] prior_lsa_next_record_internal
└─ [ 5.61%] __GI__pthread_mutex_lock
└─ [ 2.75%] __lll_lock_wait
└─ [ 4.27%] _pthread_mutex_unlock_usercnt
└─ [ 2.37%] __lll_unlock_wake
└─ [ 0.22%] prior_lsa_start_append
└─ [ 0.22%] vacuum_get_log_blockid
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 0.88%] heap_vpid_init_new
└─ [ 7.88%] log_append_undoredo_data
└─ [ 19.52%] log_append_undoredo_crumbs
└─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
└─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
└─ [ 9.96%] log_zip
└─ [ 8.84%] LZ4_compress_fast_extState
└─ [ 1.72%] LZ4_read_ARCH
└─ [ 0.78%] LZ4_read32
└─ [ 0.47%] LZ4_initStream
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.21%] LZ4_writeLE16
└─ [ 1.06%] LZ4_resetStream_fast
└─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI__libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI__mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI__libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc
└─ [ 3.59%] __GI__mprotect
└─ [ 0.68%] malloc_consolidate
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
└─ [ 6.08%] cub_alloc
└─ [ 7.24%] __GI__libc_malloc
└─ [ 6.52%] _int_malloc
└─ [ 4.20%] sysmalloc

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```

└─ [ 3.59%] __GI___mprotect
    └─ [ 0.68%] malloc_consolidate
        └─ [ 0.32%] unlink_chunk.isra.2
            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
    └─ [ 0.24%] LOG_FIND_CURRENT_TDES
        └─ [ 1.90%] LOG_FIND_TDES
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI___libc_malloc
        └─ [ 6.52%] _int_malloc
            └─ [ 4.20%] sysmalloc
                └─ [ 3.59%] __GI___mprotect
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
                            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 9.96%] log_zip
    └─ [ 8.84%] LZ4_compress_fast_extState
        └─ [ 1.72%] LZ4_read_ARCH
            └─ [ 0.78%] LZ4_read32
                └─ [ 0.47%] LZ4_initStream
                    └─ [ 1.36%] __memset_evex_unaligned_erms
                        └─ [ 0.21%] LZ4_writeLE16
└─ [ 1.06%] LZ4_resetStream_fast
    └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 0.59%] prior_lsa_copy_undo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
                                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.40%] prior_lsa_copy_redo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
                                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
    └─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 7.94%] prior_lsa_next_record_internal
    └─ [ 5.61%] __GI___pthread_mutex_lock
        └─ [ 2.75%] __lll_lock_wait
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
    └─ [ 2.37%] __lll_unlock_wake
└─ [ 0.22%] prior_lsa_start_append
    └─ [ 0.22%] vacuum_get_log_blockid
        └─ [ 1.90%] LOG_FIND_TDES
└─ [ 0.23%] spage_insert
    └─ [ 0.21%] pgbuf_set_dirty
└─ [ 0.64%] log_sysop_end_logical_undo
    └─ [ 1.17%] log_sysop_commit_internal
        └─ [ 0.99%] log_append_sysop_end

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```

└─ [ 7.94%] prior_lsa_next_record_internal
    │   └─ [ 5.61%] __GI___pthread_mutex_lock
    │       └─ [ 2.75%] __lll_lock_wait
    │   └─ [ 4.27%] __pthread_mutex_unlock_usercnt
    │       └─ [ 2.37%] __lll_unlock_wake
    │   └─ [ 0.22%] prior_lsa_start_append
    │       └─ [ 0.22%] vacuum_get_log_blockid
    └─ [ 0.45%] prior_lsa_alloc_and_copy_data
        │   └─ [ 6.08%] cub_alloc
        │       └─ [ 7.24%] __GI___libc_malloc
        │           └─ [ 6.52%] _int_malloc
        │               └─ [ 4.20%] sysmalloc
        │                   └─ [ 3.59%] __GI___mprotect
        │                       └─ [ 0.68%] malloc_consolidate
        │                           └─ [ 0.32%] unlink_chunk.isra.2
        │                               └─ [ 0.32%] unlink_chunk.isra.2
        └─ [ 0.59%] prior_lsa_copy_undo_data_to_node
            └─ [ 6.08%] cub_alloc
                └─ [ 7.24%] __GI___libc_malloc
                    └─ [ 6.52%] _int_malloc
                        └─ [ 4.20%] sysmalloc
                            └─ [ 3.59%] __GI___mprotect
                                └─ [ 0.68%] malloc_consolidate
                                    └─ [ 0.32%] unlink_chunk.isra.2
                                        └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.39%] log_tdes::unlock_topop
    └─ [ 0.38%] cubpl::get_session
        │   └─ [ 0.24%] session_get_pl_session
        │   └─ [ 0.27%] thread_get_thread_entry_info
        │       └─ [ 0.21%] cubthread::get_entry
        └─ [ 7.94%] prior_lsa_next_record_internal
            │   └─ [ 5.61%] __GI___pthread_mutex_lock
            │       └─ [ 2.75%] __lll_lock_wait
            │   └─ [ 4.27%] __pthread_mutex_unlock_usercnt
            │       └─ [ 2.37%] __lll_unlock_wake
            │   └─ [ 0.22%] prior_lsa_start_append
            │       └─ [ 0.22%] vacuum_get_log_blockid
            └─ [ 1.99%] pgbuf_unix
                │   └─ [ 1.23%] pgbuf_unlatch_bcb_upon_unix
                │       └─ [ 4.27%] __pthread_mutex_unlock_usercnt
                │           └─ [ 2.37%] __lll_unlock_wake
                └─ [ 5.61%] __GI___pthread_mutex_lock
                    └─ [ 2.75%] __lll_lock_wait
└─ [ 7.88%] log_append_undoredo_data
    └─ [ 19.52%] log_append_undoredo_crumbs
        │   └─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
        │       └─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
        │           └─ [ 9.96%] log_zip
        │               └─ [ 8.84%] LZ4_compress_fast_extState
        │                   └─ [ 1.72%] LZ4_read_ARCH
        │                       └─ [ 0.78%] LZ4_read32
        │                           └─ [ 0.47%] LZ4_initStream
        │                               └─ [ 1.36%] __memset_evex_unaligned_erms
        │                                   └─ [ 0.21%] LZ4_writeLE16
        │                                       └─ [ 1.06%] LZ4_resetStream_fast
        │                                           └─ [ 1.36%] __memset_evex_unaligned_erms
        └─ [ 6.08%] cub_alloc

```

```

└─ [ 7.24%] __GI___libc_malloc
    └─ [ 6.52%] _int_malloc
        ├── [ 4.20%] sysmalloc
        │   └─ [ 3.59%] __GI___mprotect
        ├── [ 0.68%] malloc_consolidate
        │   └─ [ 0.32%] unlink_chunk.isra.2
        └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                ├── [ 4.20%] sysmalloc
                │   └─ [ 3.59%] __GI___mprotect
                ├── [ 0.68%] malloc_consolidate
                │   └─ [ 0.32%] unlink_chunk.isra.2
                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                ├── [ 4.20%] sysmalloc
                │   └─ [ 3.59%] __GI___mprotect
                ├── [ 0.68%] malloc_consolidate
                │   └─ [ 0.32%] unlink_chunk.isra.2
                └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 0.24%] LOG_FIND_CURRENT_TDES
    └─ [ 1.90%] LOG_FIND_TDES
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI___libc_malloc
        └─ [ 6.52%] _int_malloc
            ├── [ 4.20%] sysmalloc
            │   └─ [ 3.59%] __GI___mprotect
            ├── [ 0.68%] malloc_consolidate
            │   └─ [ 0.32%] unlink_chunk.isra.2
            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 9.96%] log_zip
    ├── [ 8.84%] LZ4_compress_fast_extState
    │   ├── [ 1.72%] LZ4_read_ARCH
    │   ├── [ 0.78%] LZ4_read32
    │   ├── [ 0.47%] LZ4_initStream
    │   │   └─ [ 1.36%] __memset_evex_unaligned_erms
    │   └─ [ 0.21%] LZ4_writeLE16
    ├── [ 1.06%] LZ4_resetStream_fast
    │   └─ [ 1.36%] __memset_evex_unaligned_erms
    └─ [ 0.59%] prior_lsa_copy_undo_data_to_node
        └─ [ 6.08%] cub_alloc
            └─ [ 7.24%] __GI___libc_malloc
                └─ [ 6.52%] _int_malloc
                    ├── [ 4.20%] sysmalloc
                    │   └─ [ 3.59%] __GI___mprotect
                    ├── [ 0.68%] malloc_consolidate
                    │   └─ [ 0.32%] unlink_chunk.isra.2
                    └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.40%] prior_lsa_copy_redo_data_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc

```

```

└ [ 6.52%] _int_malloc
  └ [ 4.20%] sysmalloc
    └ [ 3.59%] __GI___mprotect
      └ [ 0.68%] malloc_consolidate
        └ [ 0.32%] unlink_chunk.isra.2
          └ [ 0.32%] unlink_chunk.isra.2
            └ [ 3.69%] __memmove_evex_unaligned_erms
              └ [ 3.69%] __memmove_evex_unaligned_erms
                └ [ 7.94%] prior_lsa_next_record_internal
                  └ [ 5.61%] __GI___pthread_mutex_lock
                    └ [ 2.75%] __lll_lock_wait
                      └ [ 4.27%] __pthread_mutex_unlock_usercnt
                        └ [ 2.37%] __lll_unlock_wake
                          └ [ 0.22%] prior_lsa_start_append
                            └ [ 0.22%] vacuum_get_log_blockid
                              └ [ 1.90%] LOG_FIND_TDES
                                └ [ 0.52%] log_sysop_commit
                                  └ [ 1.17%] log_sysop_commit_internal
                                    └ [ 0.99%] log_append_sysop_end
                                      └ [ 7.94%] prior_lsa_next_record_internal
                                        └ [ 5.61%] __GI___pthread_mutex_lock
                                          └ [ 2.75%] __lll_lock_wait
                                            └ [ 4.27%] __pthread_mutex_unlock_usercnt
                                              └ [ 2.37%] __lll_unlock_wake
                                                └ [ 0.22%] prior_lsa_start_append
                                                  └ [ 0.22%] vacuum_get_log_blockid
                                                    └ [ 0.45%] prior_lsa_alloc_and_copy_data
                                                      └ [ 6.08%] cub_alloc
                                                        └ [ 7.24%] __GI___libc_malloc
                                                          └ [ 6.52%] _int_malloc
                                                            └ [ 4.20%] sysmalloc
                                                              └ [ 3.59%] __GI___mprotect
                                                                └ [ 0.68%] malloc_consolidate
                                                                  └ [ 0.32%] unlink_chunk.isra.2
                                                                    └ [ 0.32%] unlink_chunk.isra.2
                                                                      └ [ 0.59%] prior_lsa_copy_undo_data_to_node
                                                                        └ [ 6.08%] cub_alloc
                                                                          └ [ 7.24%] __GI___libc_malloc
                                                                            └ [ 6.52%] _int_malloc
                                                                              └ [ 4.20%] sysmalloc
                                                                                └ [ 3.59%] __GI___mprotect
                                                                                  └ [ 0.68%] malloc_consolidate
                                                                                    └ [ 0.32%] unlink_chunk.isra.2
                                                                                      └ [ 0.32%] unlink_chunk.isra.2
                                                                                        └ [ 0.39%] log_tdes::unlock_topop
                                                                                          └ [ 0.38%] cubpl::get_session
                                                                                            └ [ 0.24%] session_get_pl_session
                                                                                              └ [ 0.27%] thread_get_thread_entry_info
                                                                                                └ [ 0.21%] cubthread::get_entry
                                                                                                  └ [ 0.57%] heap_stats_add_bestspace
                                                                                                    └ [ 0.34%] mht_get
                                                                                                      └ [ 5.61%] __GI___pthread_mutex_lock
                                                                                                        └ [ 2.75%] __lll_lock_wait
                                                                                                          └ [ 4.27%] __pthread_mutex_unlock_usercnt
                                                                                                            └ [ 2.37%] __lll_unlock_wake
                                                                                                              └ [ 1.99%] pgbuf_unfix
                                                                                                                └ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix

```

```

└─ [ 4.27%] __pthread_mutex_unlock_usercnt
    └─ [ 2.37%] __l1ll_unlock_wake
└─ [ 5.61%] __GI___pthread_mutex_lock
    └─ [ 2.75%] __l1ll_lock_wait
└─ [ 0.61%] log_sysop_start
    └─ [ 0.22%] log_tdes::lock_topop
        └─ [ 0.38%] cubpl::get_session
            └─ [ 0.24%] session_get_pl_session
                └─ [ 0.27%] thread_get_thread_entry_info
                    └─ [ 0.21%] cubthread::get_entry
└─ [ 0.69%] rmutex_lock
    └─ [ 5.61%] __GI___pthread_mutex_lock
        └─ [ 2.75%] __l1ll_lock_wait
            └─ [ 0.27%] thread_get_thread_entry_info
                └─ [ 0.21%] cubthread::get_entry
└─ [ 1.90%] LOG_FIND_TDES
└─ [ 7.88%] log_append_undoredo_data
    └─ [ 19.52%] log_append_undoredo_crumbs
        └─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
            └─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
                └─ [ 9.96%] log_zip
                    └─ [ 8.84%] LZ4_compress_fast_extState
                        └─ [ 1.72%] LZ4_read_ARCH
                            └─ [ 0.78%] LZ4_read32
                                └─ [ 0.47%] LZ4_initStream
                                    └─ [ 1.36%] __memset_evex_unaligned_erms
                                        └─ [ 0.21%] LZ4_writeLE16
└─ [ 1.06%] LZ4_resetStream_fast
    └─ [ 1.36%] __memset_evex_unaligned_erms
└─ [ 6.08%] cub_alloc
    └─ [ 7.24%] __GI___libc_malloc
        └─ [ 6.52%] _int_malloc
            └─ [ 4.20%] sysmalloc
                └─ [ 3.59%] __GI___mprotect
                    └─ [ 0.68%] malloc_consolidate
                        └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
    └─ [ 6.08%] cub_alloc
        └─ [ 7.24%] __GI___libc_malloc
            └─ [ 6.52%] _int_malloc
                └─ [ 4.20%] sysmalloc
                    └─ [ 3.59%] __GI___mprotect
                        └─ [ 0.68%] malloc_consolidate
                            └─ [ 0.32%] unlink_chunk.isra.2
└─ [ 0.32%] unlink_chunk.isra.2
└─ [ 3.69%] __memmove_evex_unaligned_erms
└─ [ 0.24%] LOG_FIND_CURRENT_TDES

```

```

└ [ 1.90%] LOG_FIND_TDES
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI___libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI___mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 9.96%] log_zip
└ [ 8.84%] LZ4_compress_fast_extState
└ [ 1.72%] LZ4_read_ARCH
└ [ 0.78%] LZ4_read32
└ [ 0.47%] LZ4_initStream
└ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.21%] LZ4_writeLE16
└ [ 1.06%] LZ4_resetStream_fast
└ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.59%] prior_lsa_copy_undo_data_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI___libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI___mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.40%] prior_lsa_copy_redo_data_to_node
└ [ 6.08%] cub_alloc
└ [ 7.24%] __GI___libc_malloc
└ [ 6.52%] _int_malloc
└ [ 4.20%] sysmalloc
└ [ 3.59%] __GI___mprotect
└ [ 0.68%] malloc_consolidate
└ [ 0.32%] unlink_chunk.isra.2
└ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 7.94%] prior_lsa_next_record_internal
└ [ 5.61%] __GI___pthread_mutex_lock
└ [ 2.75%] __lll_lock_wait
└ [ 4.27%] __pthread_mutex_unlock_usercnt
└ [ 2.37%] __lll_unlock_wake
└ [ 0.22%] prior_lsa_start_append
└ [ 0.22%] vacuum_get_log_blockid
└ [ 1.90%] LOG_FIND_TDES
└ [ 1.37%] heap_stats_find_page_in_bestspace
└ [ 0.57%] heap_stats_add_bestspace
└ [ 0.34%] mht_get
└ [ 5.61%] __GI___pthread_mutex_lock
└ [ 2.75%] __lll_lock_wait
└ [ 4.27%] __pthread_mutex_unlock_usercnt
└ [ 2.37%] __lll_unlock_wake
└ [ 0.30%] spage_max_space_for_new_record
└ [ 0.60%] spage_get_total_saved_spaces
└ [ 0.58%] spage_get_saved_spaces
└ [ 0.45%] lf_hash_find

```



```

└─ [ 0.26%] lf_list_find
    └─ [ 0.47%] spage_has_enough_total_space
└─ [ 5.61%] __GI___pthread_mutex_lock
    └─ [ 2.75%] __lll_lock_wait
└─ [ 4.27%] __pthread_mutex_unlock_usercnt
    └─ [ 2.37%] __lll_unlock_wake
└─ [ 1.99%] pgbuf_unfix
    └─ [ 1.23%] pgbuf unlatch_bcb_upon_unfix
        └─ [ 4.27%] __pthread_mutex_unlock_usercnt
            └─ [ 2.37%] __lll_unlock_wake
    └─ [ 5.61%] __GI___pthread_mutex_lock
        └─ [ 2.75%] __lll_lock_wait
└─ [ 0.46%] spage_get_record
    └─ [ 0.32%] spage_find_slot
└─ [ 1.75%] pgbuf_ordered_fix_release
    └─ [ 3.91%] pgbuf_fix_release
        └─ [ 1.56%] pgbuf_claim_bcb_for_fix
            └─ [ 1.33%] fileio_init_lsa_of_page
                └─ [ 1.35%] LSA_SET_NULL
        └─ [ 0.39%] pgbuf_search_hash_chain
            └─ [ 0.40%] __GI___pthread_mutex_trylock
        └─ [ 0.30%] pgbufLatch_bcb_upon_fix
        └─ [ 4.27%] __pthread_mutex_unlock_usercnt
            └─ [ 2.37%] __lll_unlock_wake
└─ [ 1.20%] btree_split_node
    └─ [ 3.68%] log_append_undo_data2
        └─ [ 3.66%] logAppend_undo_crumbs
            └─ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
                └─ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
                    └─ [ 9.96%] log_zip
                        └─ [ 8.84%] LZ4_compress_fast_extState
                            └─ [ 1.72%] LZ4_read_ARCH
                                └─ [ 0.78%] LZ4_read32
                                    └─ [ 0.47%] LZ4_initStream
                                        └─ [ 1.36%] __memset_evex_unaligned_erms
                                            └─ [ 0.21%] LZ4_writeLE16
                                                └─ [ 1.06%] LZ4_resetStream_fast
                                                    └─ [ 1.36%] __memset_evex_unaligned_erms
                └─ [ 6.08%] cub_alloc
                    └─ [ 7.24%] __GI___libc_malloc
                        └─ [ 6.52%] _int_malloc
                            └─ [ 4.20%] sysmalloc
                                └─ [ 3.59%] __GI__mprotect
                                    └─ [ 0.68%] malloc_consolidate
                                        └─ [ 0.32%] unlink_chunk.isra.2
                                            └─ [ 0.32%] unlink_chunk.isra.2
                └─ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
                    └─ [ 6.08%] cub_alloc
                        └─ [ 7.24%] __GI___libc_malloc
                            └─ [ 6.52%] _int_malloc
                                └─ [ 4.20%] sysmalloc
                                    └─ [ 3.59%] __GI__mprotect
                                        └─ [ 0.68%] malloc_consolidate
                                            └─ [ 0.32%] unlink_chunk.isra.2
                                                └─ [ 0.32%] unlink_chunk.isra.2
                    └─ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
                        └─ [ 6.08%] cub alloc
```

```

└ [ 7.24%] __GI___libc_malloc
  └ [ 6.52%] _int_malloc
    └ [ 4.20%] sysmalloc
      └ [ 3.59%] __GI___mprotect
        └ [ 0.68%] malloc_consolidate
          └ [ 0.32%] unlink_chunk.isra.2
            └ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 0.24%] LOG_FIND_CURRENT_TDES
  └ [ 1.90%] LOG_FIND_TDES
└ [ 6.08%] cub_alloc
  └ [ 7.24%] __GI___libc_malloc
    └ [ 6.52%] _int_malloc
      └ [ 4.20%] sysmalloc
        └ [ 3.59%] __GI___mprotect
          └ [ 0.68%] malloc_consolidate
            └ [ 0.32%] unlink_chunk.isra.2
              └ [ 0.32%] unlink_chunk.isra.2
└ [ 9.96%] log_zip
  └ [ 8.84%] LZ4_compress_fast_extState
    └ [ 1.72%] LZ4_read_ARCH
      └ [ 0.78%] LZ4_read32
        └ [ 0.47%] LZ4_initStream
          └ [ 1.36%] __memset_evex_unaligned_erms
            └ [ 0.21%] LZ4_writeLE16
              └ [ 1.06%] LZ4_resetStream_fast
                └ [ 1.36%] __memset_evex_unaligned_erms
└ [ 0.59%] prior_lsa_copy_undo_data_to_node
  └ [ 6.08%] cub_alloc
    └ [ 7.24%] __GI___libc_malloc
      └ [ 6.52%] _int_malloc
        └ [ 4.20%] sysmalloc
          └ [ 3.59%] __GI___mprotect
            └ [ 0.68%] malloc_consolidate
              └ [ 0.32%] unlink_chunk.isra.2
                └ [ 0.32%] unlink_chunk.isra.2
└ [ 0.40%] prior_lsa_copy_redo_data_to_node
  └ [ 6.08%] cub_alloc
    └ [ 7.24%] __GI___libc_malloc
      └ [ 6.52%] _int_malloc
        └ [ 4.20%] sysmalloc
          └ [ 3.59%] __GI___mprotect
            └ [ 0.68%] malloc_consolidate
              └ [ 0.32%] unlink_chunk.isra.2
                └ [ 0.32%] unlink_chunk.isra.2
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 3.69%] __memmove_evex_unaligned_erms
└ [ 7.94%] prior_lsa_next_record_internal
  └ [ 5.61%] __GI___pthread_mutex_lock
    └ [ 2.75%] __lll_lock_wait
      └ [ 4.27%] __pthread_mutex_unlock_usercnt
        └ [ 2.37%] __lll_unlock_wake
└ [ 0.22%] prior_lsa_start_append
  └ [ 0.22%] vacuum_get_log_blockid
└ [ 1.90%] LOG_FIND_TDES
└ [ 0.37%] log_append_redo_data2
  └ [ 0.37%] log_append_redo_crumbs

```

```

└ [ 15.54%] prior_lsa_alloc_and_copy_crumbs
  └ [ 11.92%] prior_lsa_gen_undoredo_record_from_crumbs
    └ [ 9.96%] log_zip
      └ [ 8.84%] LZ4_compress_fast_extState
        └ [ 1.72%] LZ4_read_ARCH
          └ [ 0.78%] LZ4_read32
            └ [ 0.47%] LZ4_initStream
              └ [ 1.36%] __memset_evex_unaligned_erms
                └ [ 0.21%] LZ4_writeLE16
                  └ [ 1.06%] LZ4_resetStream_fast
                    └ [ 1.36%] __memset_evex_unaligned_erms
  └ [ 6.08%] cub_alloc
    └ [ 7.24%] __GI__libc_malloc
      └ [ 6.52%] _int_malloc
        └ [ 4.20%] sysmalloc
          └ [ 3.59%] __GI__mprotect
            └ [ 0.68%] malloc_consolidate
              └ [ 0.32%] unlink_chunk.isra.2
                └ [ 0.32%] unlink_chunk.isra.2
  └ [ 0.53%] prior_lsa_copy_undo_crumbs_to_node
    └ [ 6.08%] cub_alloc
      └ [ 7.24%] __GI__libc_malloc
        └ [ 6.52%] _int_malloc
          └ [ 4.20%] sysmalloc
            └ [ 3.59%] __GI__mprotect
              └ [ 0.68%] malloc_consolidate
                └ [ 0.32%] unlink_chunk.isra.2
                  └ [ 0.32%] unlink_chunk.isra.2
  └ [ 0.32%] prior_lsa_copy_redo_crumbs_to_node
    └ [ 6.08%] cub_alloc
      └ [ 7.24%] __GI__libc_malloc
        └ [ 6.52%] _int_malloc
          └ [ 4.20%] sysmalloc
            └ [ 3.59%] __GI__mprotect
              └ [ 0.68%] malloc_consolidate
                └ [ 0.32%] unlink_chunk.isra.2
                  └ [ 0.32%] unlink_chunk.isra.2
  └ [ 3.69%] __memmove_evex_unaligned_erms
  └ [ 0.24%] LOG_FIND_CURRENT_TDES
    └ [ 1.90%] LOG_FIND_TDES
  └ [ 6.08%] cub_alloc
    └ [ 7.24%] __GI__libc_malloc
      └ [ 6.52%] _int_malloc
        └ [ 4.20%] sysmalloc
          └ [ 3.59%] __GI__mprotect
            └ [ 0.68%] malloc_consolidate
              └ [ 0.32%] unlink_chunk.isra.2
                └ [ 0.32%] unlink_chunk.isra.2
  └ [ 9.96%] log_zip
    └ [ 8.84%] LZ4_compress_fast_extState
      └ [ 1.72%] LZ4_read_ARCH
        └ [ 0.78%] LZ4_read32
          └ [ 0.47%] LZ4_initStream
            └ [ 1.36%] __memset_evex_unaligned_erms
              └ [ 0.21%] LZ4_writeLE16
                └ [ 1.06%] LZ4_resetStream_fast
                  └ [ 1.36%] __memset_evex_unaligned_erms

```

```
tbb::detail::d2::internal_try_pop_impl<tbb::detail::d2::concurrent_queue_rep<subconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<subconn::connection::worker::message> >,
tbb::detail::d1::cache_aligned_allocator<tbb::detail::d2::concurrent_queue_rep<subconn::connection::worker::message
, tbb::detail::d1::cache_aligned_allocator<subconn::connection::worker::message> > > >
```

```
|      |   |   |       └ [ 0.23%] cub_free  
|      |   |   |           └ [ 0.33%] _int_free  
|      |   |   |       └ [ 0.85%] cubconn::connection::worker::purge_stale_contexts  
|      |   |   |           ├── [ 0.52%] cubconn::connection::coordinator::notify  
|      |   |   |               │ └ [ 0.50%] __libc_write  
|      |   |   |           └ [ 0.34%] cubconn::connection::coordinator::enqueue  
|      |   |   |       └ [ 0.22%] tbb::detail::d2::micro_queue<cubconn::connection::coordinator::message,  
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::coordinator::message> >::prepare_page(unsigned long,  
tbb::detail::d2::concurrent_queue_rep<cubconn::connection::coordinator::message,  
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::coordinator::message> >&  
tbb::detail::d1::cache_aligned_allocator<tbb::detail::d2::micro_queue<cubconn::connection::coordinator::message,  
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::coordinator::message> >::padded_page>,  
tbb::detail::d2::micro_queue<cubconn::connection::coordinator::message,  
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::coordinator::message> >::padded_page*&)::  
{lambda(#1)}::operator()  
|      |   |   |           └ [ 0.22%] tbb::detail::r1::cache_aligned_allocate  
|      |   |   |       └ [ 0.45%] cubconn::connection::worker::eventfd_clear  
|      |   |   |           └ [ 0.43%] __libc_read  
|      |   |   |       ├── [ 3.91%] cubsocket::epoll::wait  
|      |   |   |           │ └ [ 3.88%] epoll_wait  
|      |   |   |       └ [ 3.86%] cubconn::connection::worker::handle_reception  
|      |   |   |           ├── [ 2.31%] cubconn::receiver::drain  
|      |   |   |               │ └ [ 2.15%] cubconn::receiver::receive  
|      |   |   |                   │ └ [ 2.13%] __libc_recv  
|      |   |   |           ├── [ 0.86%] cubconn::connection::worker::handle_data_packet  
|      |   |   |               │ └ [ 0.57%] cubthread::worker_pool::core::execute_task  
|      |   |   |                   │ └ [ 0.48%] cubthread::worker_pool::core::worker::assign_task  
|      |   |   |                       │ └ [ 0.45%] std::condition_variable::notify_one  
|      |   |   |                           │ └ [ 0.44%] __pthread_cond_signal  
|      |   |   |           ├── [ 0.21%] cubconn::connection::worker::handle_header_packet  
|      |   |   |               │ └ [ 0.55%] cubconn::receiver::release  
|      |   |   |                   │ └ [ 0.40%] cubbase::DMRBMemoryPool::restore  
|      |   |   |           └── [ 0.22%] std::_Rb_tree<unsigned long, std::pair<unsigned long const, unsigned long>,  
std::_Select1st<std::pair<unsigned long const, unsigned long> >, std::less<unsigned long>,  
std::allocator<std::pair<unsigned long const, unsigned long> > >::_M_emplace_unique<unsigned long&, unsigned long&>  
|      |   |   |           └ [ 0.22%] operator delete  
|      |   |   |       └ [ 0.23%] cub_free  
|      |   |   |           └ [ 0.33%] _int_free  
|      |   |   |       └ [ 0.52%] rmutex_lock  
└ [ 5.33%] «vacuum» (7847 smp)  
|   ├── [ 2.33%] vacuum_process_log_block  
|   |   └ [ 2.01%] btree_vacuum_insert_mvccid  
|   |       └ [ 2.00%] btree_delete_internal  
|   |           ├── [ 1.38%] btree_merge_node_and_advance  
|   |               │ ├── [ 1.04%] pgbuf_fix_release  
|   |                   │ └ [ 0.51%] pgbuf_latch_bcb_upon_fix  
|   |                       │ └ [ 0.40%] pgbuf_block_bcb  
|   |                           │ └ [ 0.36%] thread_suspend_timeout_wakeup_and_unlock_entry  
|   |                               │ └ [ 0.33%] __pthread_cond_timedwait  
|   |                                   └ [ 0.31%] __pthread_mutex_unlock_usercnt  
|   |                                       └ [ 0.22%] __l1l_unlock_wake  
|   |           ├── [ 0.30%] btree_search_leaf_page  
|   |               │ └ [ 0.20%] btree_read_record_without_decompression  
|   |                   ├── [ 0.23%] btree_search_nonleaf_page  
|   |                       │ └ [ 0.20%] btree_read_record_without_decompression  
|   |                           └ [ 0.30%] btree_search_leaf_page  
|   |           └ [ 0.45%] pqbuf_unfix
```

```

└─ [ 0.26%] pgbuf_unlatch_bcb_upon_unfix
    └─ [ 0.31%] __pthread_mutex_unlock_usercnt
        └─ [ 0.22%] __l1l_unlock_wake
    └─ [ 0.46%] __GI__pthread_mutex_lock
        └─ [ 0.22%] __l1l_lock_wait
└─ [ 0.58%] btree_search_key_and_apply_functions
    └─ [ 0.45%] pgbuf_unfix
        └─ [ 0.26%] pgbuf_unlatch_bcb_upon_unfix
            └─ [ 0.31%] __pthread_mutex_unlock_usercnt
                └─ [ 0.22%] __l1l_unlock_wake
            └─ [ 0.46%] __GI__pthread_mutex_lock
                └─ [ 0.22%] __l1l_lock_wait
        └─ [ 0.28%] btree_fix_root_for_delete
            └─ [ 1.04%] pgbuf_fix_release
                └─ [ 0.51%] pgbuf_latch_bcb_upon_fix
                    └─ [ 0.40%] pgbuf_block_bcb
                        └─ [ 0.36%] thread_suspend_timeout_wakeup_and_unlock_entry
                            └─ [ 0.33%] __pthread_cond_timedwait
                                └─ [ 0.31%] __pthread_mutex_unlock_usercnt
                                    └─ [ 0.22%] __l1l_unlock_wake
└─ [ 1.70%] vacuum_heap_page
    └─ [ 0.86%] vacuum_heap_prepare_record
        └─ [ 0.82%] spage_get_record_data
            └─ [ 1.20%] __memmove_evex_unaligned_erms
    └─ [ 0.39%] spage_update
        └─ [ 0.29%] spage_update_record_in_place
            └─ [ 1.20%] __memmove_evex_unaligned_erms
    └─ [ 0.29%] vacuum_heap_page_log_and_reset
        └─ [ 0.26%] heap_stats_update
            └─ [ 0.26%] heap_stats_add_bestspace
                └─ [ 0.46%] __GI__pthread_mutex_lock
                    └─ [ 0.22%] __l1l_lock_wait
                └─ [ 0.31%] __pthread_mutex_unlock_usercnt
                    └─ [ 0.22%] __l1l_unlock_wake
                └─ [ 0.36%] cub_alloc
                    └─ [ 0.37%] __GI__libc_malloc
                        └─ [ 0.29%] _int_malloc
                            └─ [ 0.21%] sysmalloc
            └─ [ 0.45%] pgbuf_unfix
                └─ [ 0.26%] pgbuf_unlatch_bcb_upon_unfix
                    └─ [ 0.31%] __pthread_mutex_unlock_usercnt
                        └─ [ 0.22%] __l1l_unlock_wake
                └─ [ 0.46%] __GI__pthread_mutex_lock
                    └─ [ 0.22%] __l1l_lock_wait
        └─ [ 1.04%] pgbuf_fix_release
            └─ [ 0.51%] pgbuf_latch_bcb_upon_fix
                └─ [ 0.40%] pgbuf_block_bcb
                    └─ [ 0.36%] thread_suspend_timeout_wakeup_and_unlock_entry
                        └─ [ 0.33%] __pthread_cond_timedwait
                            └─ [ 0.31%] __pthread_mutex_unlock_usercnt
                                └─ [ 0.22%] __l1l_unlock_wake
└─ [ 1.04%] btree_key_remove_insert_mvccid
    └─ [ 1.04%] log_append_redo_data
        └─ [ 1.03%] log_append_redo_crumbs
            └─ [ 0.54%] prior_lsa_next_record_internal
                └─ [ 0.46%] __GI__pthread_mutex_lock
                    └─ [ 0.22%] __l1l_lock_wait

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|      └ [ 0.78%] __libc_pwrite64
└ [ 1.69%] «coordinator» (2488 smp)
|   └ [ 1.64%] __GI__clone
|       └ [ 1.64%] start_thread
|           └ [ 1.64%] execute_native_thread_routine
|               └ [ 1.64%] cubconn::connection::coordinator::attach
|                   └ [ 1.64%] cubconn::connection::coordinator::run
|                       └ [ 1.11%] cubsocket::epoll::wait
|                           └ [ 1.11%] epoll_wait
|                               └ [ 0.25%] cubconn::connection::coordinator::handle_message_queue
|                                   └ [ 0.20%]
tbb::detail::d2::internal_try_pop_impl<tbb::detail::d2::concurrent_queue_rep<cubconn::connection::coordinator::mess
age, tbb::detail::d1::cache_aligned_allocator<cubconn::connection::coordinator::message> >,
tbb::detail::d1::cache_aligned_allocator<tbb::detail::d2::concurrent_queue_rep<cubconn::connection::coordinator::me
ssage, tbb::detail::d1::cache_aligned_allocator<cubconn::connection::coordinator::message> > > >
|           └ [ 0.21%] cubconn::connection::coordinator::eventfd_clear
└ [ 0.85%] «vacuum-master» (1253 smp)
|   └ [ 0.81%] __GI__clone
|       └ [ 0.80%] start_thread
|           └ [ 0.80%] execute_native_thread_routine
|               └ [ 0.80%] cubthread::daemon::loop_with_context
|                   └ [ 0.64%] cubthread::looper::put_to_sleep
|                       └ [ 0.62%] cubthread::waiter::wait_for
|                           └ [ 0.53%] __pthread_cond_timedwait
└ [ 0.73%] «dwb-file-sync» (1080 smp)
|   └ [ 0.73%] __GI__clone
|       └ [ 0.73%] start_thread
|           └ [ 0.73%] execute_native_thread_routine
|               └ [ 0.73%] cubthread::daemon::loop_with_context
|                   └ [ 0.68%] cubthread::looper::put_to_sleep
|                       └ [ 0.66%] cubthread::waiter::wait_for
|                           └ [ 0.57%] __pthread_cond_timedwait
└ [ 1.05%] «기타 thread 9종» (1549 smp, 각 <0.3%)
    0.27% pgbuf-flush-con (399)
    0.25% deadlock-detect (370)
    0.22% pgbuf-maintain (324)
    0.11% log-clock (161)
    0.08% ha-delay-check (119)
    0.04% pl-monitor (61)
    0.04% cub_server (60)
    0.03% pgbuf-page-flus (46)
    0.01% session-control (9)

```